A Cross-Species Lexical Repertoire Reveals a Biological Code Linking Acoustic Form to Communicative Meaning

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Abstract

Understanding animal vocalizations means connecting acoustic form to communicative meaning. Years of field studies have produced extensive knowledge of such mappings, yet no unified framework exists for comparing them across species. We introduce AnimalLex, a collaborative, curated cross-species repertoire database comprising 373 calls from 57 species spanning mammals, birds, and amphibians (TODO: 10x), annotated with standardized semantic and acoustic descriptions synthesized from the scientific literature [and validated by domain experts/with X% agreement with domain experts]. Using AnimalLex, we characterize what animals talk about and how. We find that acoustic and semantic distances are significantly correlated within species, across related species, and even across taxa—evidence for a conserved form-to-meaning structure that transcends phylogenetic boundaries. We leverage this structure to perform cross-species lexical translation: given any call, we can identify its closest analogue in any other species. These results suggest that a biological code linking acoustic form to communicative function is a deep organizational principle of animal communication, and that large-scale repertoire curation—now made practical by language models—unlocks new possibilities for comparative and translational bioacoustics. An interactive tool is provided to explore the database and perform translations at scale.

1 Introduction

Decades of field research have established that animal vocalizations are not simple reflexes but structured communicative signals whose forms are shaped by function Marler, 1955. Alarm calls are short, broadband, and easily localized—suited for rapid threat broadcast. Contact calls are tonal and repetitive—suited for long-distance propagation and individual recognition. Infant-directed calls are high-pitched across taxa—plausibly linked to shared attentional mechanisms. Yet these patterns have been documented one species at a time, and no synthesis has asked whether they constitute a universal code: a systematic, learnable relationship between acoustic form and communicative meaning that holds across the tree of life.

The question is not merely empirical but conceptual. In linguistics, the arbitrariness of the sign—the principle that a word's sound bears no necessary relation to its meaning—has been a foundational tenet since Saussure Saussure, 1916. Counter-evidence is accumulating: sound symbolism and crosslinguistic iconicity demonstrate that form-to-meaning mappings exist and are learned in human language Dingemanse et al., 2015. In non-human animals the question is more open still. If such a biological code exists, it could arise through two mechanisms: convergent evolution driven by shared physical and ecological constraints (e.g., the localizability demands of alarm signaling), or phylogenetic conservation of ancestral form-to-meaning associations within lineages. These mechanisms make different predictions—the former produces cross-taxon universals, the latter family-level regularities—and can in principle be distinguished with sufficient data.

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Here we test for a biological code at scale by assembling AnimalLex, a curated database of 373 calls from 57 species (Fig. 1). Each entry pairs a free-text acoustic description with a free-text semantic description and standardized ontology keywords, drawn from peer-reviewed literature and validated by domain experts. We embed both descriptions independently into continuous vector spaces and ask whether their pairwise distance structures are aligned—within species, across species, and across higher taxa. We find that they are: acoustic distance is a significant predictor of semantic distance at every taxonomic level tested. A simple linear model trained on this alignment generalizes across species and taxa, demonstrating that the code is learnable. We apply it directly to perform cross-species lexical translation—mapping any documented call to its closest human-language analogue, and any human concept to its closest animal vocalization—and provide an interactive web application for broad exploration.

2 A cross-species lexical repertoire

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Figure 1. **AnimalLex: a curated cross-species lexical repertoire. (A)** Frequency of semantic ontology keywords across all 373 calls, colored by communicative category. Coordination, contact, alarm, and affective signaling dominate; rarer categories include referential signaling, individual identity, and learning-related calls. **(B)** Frequency of acoustic features extracted by term-matching from free-text descriptions. Tonal, high-frequency, and repetitive features are most common; soft/quiet and harmonic features are rarer. **(C)** Calls per species, grouped by taxonomic group (Apes, Old World monkeys, other primates, Carnivora, other mammals, Passeriformes, Psittaciformes, Amphibia) and colored by family. Within each group, species are sorted by repertoire size. Repertoire sizes range from 2 to 32; the largest is the Bornean orangutan (32 calls), reflecting the depth of primary literature on great apes.

AnimalLex was assembled by prompting a large language model (GPT-4o) to structure knowledge from peer-reviewed literature for each target species, producing entries that record call name, free-text acoustic description, free-text semantic description, standardized ontology keywords, subjective reliability, and primary references. Entries were then reviewed and corrected by domain experts (see Methods). The 57 species span three vertebrate classes: 43 mammals, 9 birds, and 5 amphibians, covering taxa from African savanna elephants (*Loxodonta africana*) and gray wolves (*Canis lupus*) to Carolina chickadees (*Poecile carolinensis*) and túngara frogs (*Engystomops pustulosus*).

The most common semantic categories are coordination ($n = 106$), affective signaling ($n = 98$), contact and cohesion ($n = 94$), distress ($n = 72$), and alarm ($n = 71$; Fig. 1A). Less frequent but theoretically important categories include referential signals ($n = 5$), individual-identity calls ($n = 8$), and learning-associated vocalizations ($n = 10$). On the acoustic side, descriptions span the full range of spectro-temporal parameters: tonal vs. broadband energy, frequency range and modulation, temporal patterning (pulsed, repetitive, graded), and call-internal structure (notes, syllables, phrases; Fig. 1B). Together these two annotation axes provide the substrate for testing form-to-meaning alignment.

3 Acoustic and semantic spaces are aligned across the tree of life

We embedded each call's acoustic and semantic descriptions separately into a shared sentence-transformer space (see Methods) and computed pairwise cosine similarities across all $373(373-1)/2 = 69{,}378$ call pairs. A Mantel test Mantel, 1967 of the correlation between acoustic and semantic similarity matrices yields $r = 0.31$, $p < 0.001$ (9,999 permutations; Fig. 2C, Fig. 3 and Supplementary Material D). A partial Mantel test controlling for species identity yields an identical coefficient ($r = 0.31$, $p < 0.001$), confirming that the correlation is not driven by the trivial within-species similarity structure: calls from the same species are not the reason acoustic and semantic spaces are aligned.

Examining subsets of pairs by taxonomic relationship reveals additional structure. Pairs from the same family but different species show the strongest correlation ($r = 0.49$, $p < 0.001$, $n = 13{,}021$ pairs), while within-species pairs (pooled across all species; $r = 0.23$, $p < 0.001$, $n = 1{,}551$) and cross-family pairs ($r = 0.23$, $p < 0.001$, $n = 54{,}806$) are weaker and similar to each other. The same-family peak suggests that acoustic-semantic alignment is most visible when species share evolutionary history but have begun to acoustically diverge—consistent with a code that is phylogenetically conserved while acoustic surface form

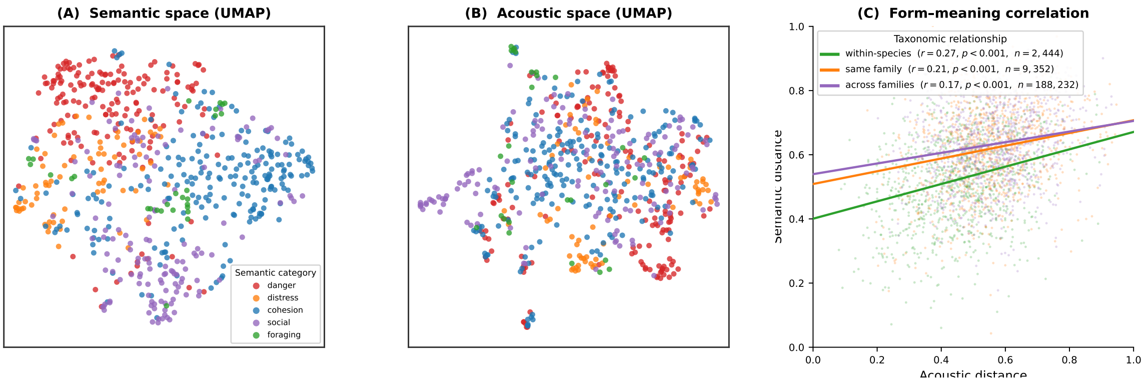

Figure 2. **Alignment of acoustic and semantic embedding spaces.** **(A)** Semantic UMAP: each point is a call, colored by communicative category (legend inside panel). Alarm/threat calls (red) cluster together; contact/cohesion calls (blue) form a separate region. **(B)** Acoustic UMAP: the same calls positioned by their acoustic description embedding. The topological similarity to (A)—shared cluster identities despite independent embeddings—is the visual signature of the biological code. **(C)** Mantel scatter: 1,000 randomly sampled pairs per group, pairwise acoustic distance vs. semantic distance, colored by taxonomic relationship. r values from permutation Mantel tests on all pairs; same-family cross-species pairs ($r = 0.49$) show the strongest alignment.

drifts. Figure 3 shows the full pairwise Mantel correlations, and reveal clusters of species with similar form-to-meaning mappings, within and across families.

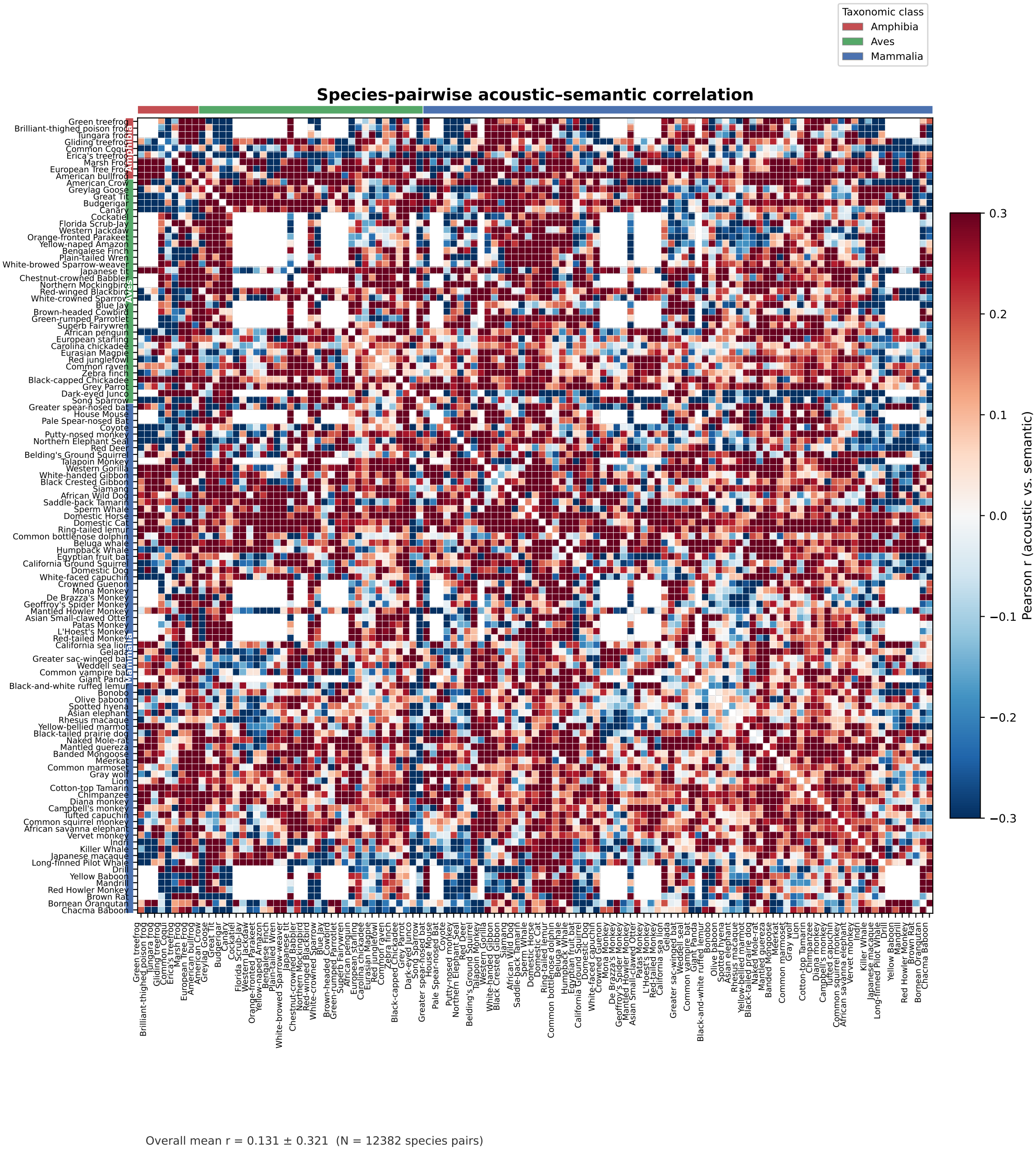

Figure 3. Species-pairwise acoustic–semantic correlation. Each cell shows the Pearson r between the vector of all acoustic-similarity values and the vector of all semantic-similarity values for every species pair. Species are sorted by taxonomic class (coloured bars: Amphibia, Aves, Mammalia), with hierarchical clustering within each class to reveal similarity structure. Positive correlations dominate within each taxonomic class and persist broadly across classes, confirming that the form-to-meaning alignment is not an artefact of pooling.

4 Form-to-meaning mapping: cracking the code?

To decompose the signal into interpretable units, we computed the pointwise mutual information (PMI) between acoustic features and semantic keywords across all calls (Fig. 4). Several strong, interpretable associations emerge. High-frequency calls are over-represented in infant-directed and food contexts (PMI ≈ 2.1–2.3) and sharply under-represented in display and aggression contexts (PMI ≈ -3.3 to -3.4). Broadband and noisy calls are enriched for aggression and threat semantics (PMI ≈ 1.7), consistent with the "harsh = aggressive" frequency code Morton, 1977. Long/sustained and loud features co-occur with long-distance cohesion calls (PMI ≈ 1.8–2.0), consistent with propagation demands. These keyword-level associations provide a transparent, mechanistic decomposition of the biological code and replicate functional acoustic ecology predictions across 57 species simultaneously.

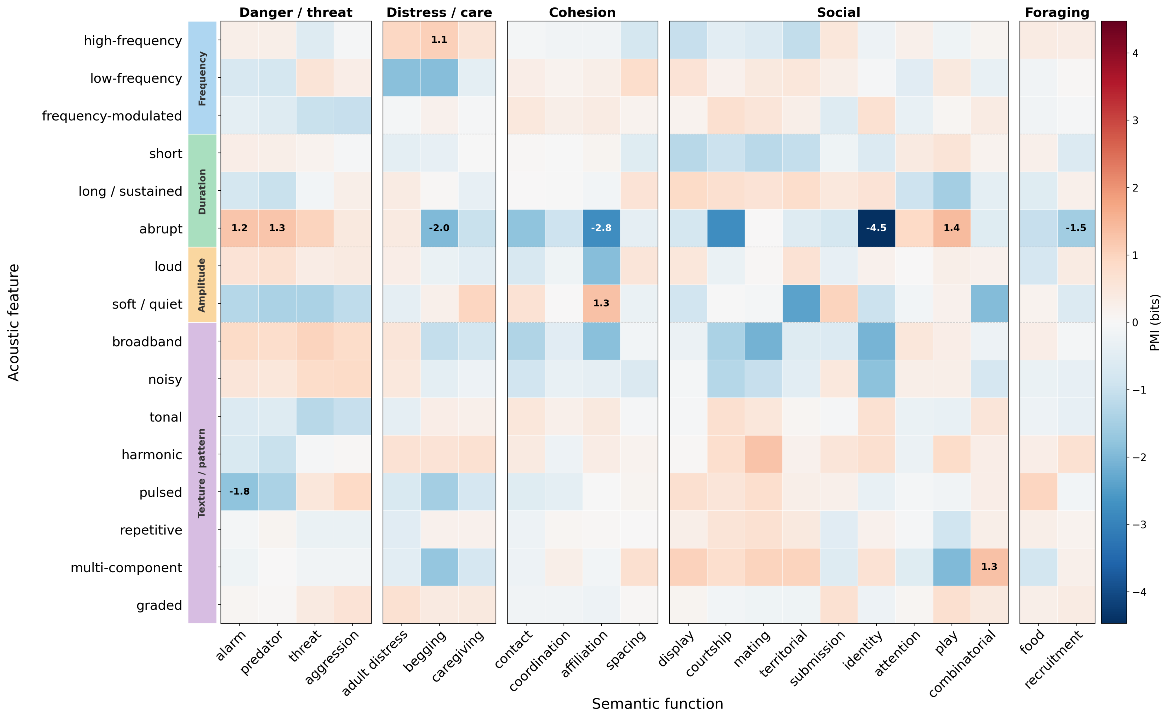

Figure 4. **Keyword-level structure of the biological code.** Pointwise mutual information (PMI, bits) between semantic functions (columns, grouped into panels by category) and acoustic features (rows, grouped by type), estimated across all 373 calls in AnimalLex. Acoustic features are ordered by group: frequency (high, low, FM), duration (short, long), amplitude (loud, soft), and texture/pattern. Numbers annotate the two strongest and one weakest association per panel. Warm colors = acoustic feature over-represented in calls with that semantic function; cool colors = under-represented.

5 Form-to-meaning mapping is learnable and generalizes across taxa

Table 1. **Cross-generalization of acoustic-to-semantic prediction.** Four models are evaluated under three hold-out conditions of increasing difficulty. *Cos.* = mean cosine similarity between predicted and true semantic embedding (higher is better, max 1). *MRR* = mean reciprocal rank of the true call among all test-set calls ranked by similarity to the prediction (higher is better, max 1). The random baseline shuffles acoustic–semantic pairings within each test set. Retrieval ($k = 1$) returns the nearest neighbour in acoustic space among training calls. Within-repertoire values are means over 10 random folds; held-out species and family values are means over leave-one-out folds (55 and 26 folds respectively).

	Within repertoire		Held-out species		Held-out family	
Model	Cos.	MRR	Cos.	MRR	Cos.	MRR
Random	0.36	0.11	0.46	0.41	0.47	0.42
Retrieval ($k = 1$)	0.67	**0.51**	0.68	0.78	0.60	0.67
Linear (ridge)	**0.74**	0.46	**0.74**	**0.80**	**0.68**	**0.70**
MLP	0.71	0.45	0.73	**0.80**	0.66	0.68

The previous section demonstrates that there is stability in the form-to-meaning mapping. We trained models to exploit this regularity: predicting semantic embeddings from acoustic embeddings under three hold-out conditions of increasing ambition (Table 1). Within-repertoire, ridge regression achieves a mean cosine similarity of 0.74 and MRR of 0.46—substantially above the random baseline (cos. 0.36, MRR 0.11). Generalization to held-out species is equally strong (cos. 0.74, MRR 0.80), and remains well above chance even when entire families are held out (cos. 0.68, MRR 0.70). The retrieval baseline approaches ridge regression in MRR for held-out conditions, indicating that the acoustic-to-semantic mapping is largely a global alignment of the two spaces rather than a complex nonlinear transformation. The MLP offers no consistent advantage over ridge regression, consistent with this interpretation. Together these results show that the biological code identified by the Mantel analysis is not an artefact of within-species structure: it generalises to species and entire families unseen at training time. In other words, our existing knowledge of species transfers to new species.

6 Lexical translation across species

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Figure 5. **Cross-species call groups sharing acoustic and semantic form.** Each panel shows one group of calls—one per species—belonging to a single semantic category (dark header). A pair of calls qualifies iff each is simultaneously the top-1 acoustic *and* top-1 semantic nearest neighbour of the other within its counterpart species; groups are cliques in this species-pairwise mutual-consistency graph, selected to maximise taxonomic diversity (class and family spread; see Supplementary Sections C and G). The coloured dot encodes taxonomic class (legend). *Top:* 2×2 layout; each row shows species and call name, followed by semantic (*Sem:*) and acoustic (*Acc:*) descriptions. *Bottom:* same four groups arranged horizontally; each row shows common name (line 1), call name (line 2), semantic keywords (line 3), and acoustic description in italics (line 4).

AnimalLex allows best guess translations across the animal kingdom. Given a call from species A, we retrieve the call in species B whose semantic embedding is nearest—a lexical translation from one repertoire to another. When both species are represented in AnimalLex with full semantic annotations, the mapping is performed in semantic space by retrieval. Given the form-to-meaning structure identified above, translational attempts are also possible on the basis of acoustic annotations only—as will routinely be the case for poorly studied taxa—the acoustic-to-semantic predictor from the previous section supplies a best-guess semantic embedding from acoustic features alone, enabling translation even across the boundary of documented knowledge.

This two-mode pipeline (retrieval for annotated calls; prediction for unannotated ones) makes AnimalLex a living resource: every new species added expands the reach of translation, and the predictor provides a scaffold for annotating the next. We release an interactive web application (https://emmanuel-dev.tail3ea00b.ts.net/) in which users can select any call and browse its nearest semantic neighbours across all species in AnimalLex, with the option to include predicted-semantics calls from unannotated recordings (see Appendix A). Strikingly, Fig. 5 shows examples where several calls can be closest in semantic space from each other as well as closest in acoustic space across broad families of species; this confirms the form-to-meaning stability.

A natural extension is to learn the translation *directly in acoustic space*, bypassing the semantic intermediate. Two species that are phylogenetically close should have nearly aligned acoustic spaces—similar vocal anatomies, shared evolutionary pressures, conserved call repertoires—and their repertoires should be mappable by a small rotation. Phylogenetically distant species may require a larger transformation. This suggests learning a family of species-pair rotations R_{AB} (Procrustes-style, constrained to be orthogonal) regularised by phylogenetic distance: $\|R_{AB} - \mathbf{I}\|_F \leq \lambda \cdot d_{\text{phylo}}(A, B)$, where d_{phylo} is branch length in a dated phylogeny and λ controls how much divergence is allowed per unit evolutionary distance. Such a model would be fully audio-grounded, require no semantic labels, and could scale to the thousands of species for which recordings but no annotations exist. We leave this to future work, but note that the form-to-meaning results above provide the theoretical justification: because acoustic and semantic spaces are aligned, a rotation in acoustic space is also, approximately, a rotation in semantic space.

7 Discussion

Our results establish three things. First, animal vocalizations carry systematic form-to-meaning relationships detectable across the tree of life—not just within well-studied model systems. Second, these relationships are learnable and generalizable: a model trained on some species predicts communicative meaning from acoustics for entirely held-out species and families. Third, this learnability enables practical cross-species lexical translation—finding, for any call, its closest semantic analogue in any other repertoire—including for species whose calls have never been semantically annotated.

Two non-exclusive mechanisms can explain the cross-taxon alignment. Convergent functional pressures may independently channel evolution toward similar acoustic solutions for similar communicative problems: the physics of sound localization favors broadband calls for alarm regardless of ancestry Marler, 1955; the acoustics of long-range propagation favor tonality for cohesion calls Morton, 1977. Phylogenetic inertia may additionally preserve ancestral form-to-meaning associations within lineages, explaining why correlations are strongest within species and weaken—without vanishing—across families. These mechanisms predict different patterns in phylogenetically controlled analyses, which future work with broader taxonomic sampling can test.

Several caveats apply. AnimalLex represents the current state of published knowledge, which skews toward well-studied mammals; birds and amphibians—acoustically diverse and experimentally tractable—are underrepresented. Our acoustic embeddings are derived from textual descriptions, not from audio directly; future work integrating audio representations Hagiwara, 2022; Robinson et al., 2025 will test whether the form-to-meaning alignment is stronger in the raw signal domain. Lexical translation should be understood as semantic proximity within the space of documented ethological descriptions—not as a claim about subjective experience, cognitive content, or the *Umwelt* of the signaling animal. Any mapping through human language (see Appendix B) inherits an additional anthropomorphic bias and should be treated as a heuristic rather than a literal gloss.

Despite these limitations, the demonstration that form-to-meaning structure is learnable and cross-taxon generalizable opens a practical path forward: with each new species added to AnimalLex, the translation model improves, and with each improvement the model can bootstrap annotation of the next species. This virtuous cycle—human knowledge, compressed by language models, refined by acoustic-semantic prediction, extended to undocumented taxa—is the long-term promise of the approach introduced here.

8 Methods

Database construction AnimalLex was assembled in two stages. We first prompted GPT-4o OpenAI, 2024 with a structured schema specifying six required fields per call: (1) call name, (2) free-text acoustic description (spectro-temporal features, amplitude envelope, call-internal structure), (3) free-text semantic description (communicative function, behavioral context, receiver responses), (4) ontology keywords drawn from a fixed controlled vocabulary, (5) subjective reliability rating (high / medium / low / contested), and (6) URLs to primary-literature references. The model was instructed to restrict outputs to findings supported by peer-reviewed sources and to flag contested or uncertain claims. In the second stage, domain experts in avian, mammalian, and amphibian bioacoustics reviewed all entries, corrected factual errors, added missing calls, and updated reliability ratings. Calls rated "low" or "contested" are retained in AnimalLex but excluded from quantitative analyses reported here.

Embeddings Acoustic and semantic descriptions were embedded independently using the all-MiniLM-L6-v2 sentence transformer Reimers and Gurevych, 2019, which maps variable-length text to a 384-dimensional vector space trained on semantic similarity tasks. Embeddings were ℓ_2-normalized before computing cosine distances. We deliberately used textual embeddings rather than audio-derived representations to ensure that acoustic and semantic descriptions were embedded in a comparable space; the validity of this choice is discussed above and tested in Supplementary E.

Mantel test For each subset of calls (all pairs; within-species pairs; within-family pairs; cross-family pairs), we computed the Pearson correlation between the upper triangle of the pairwise acoustic distance matrix and the upper triangle of the pairwise semantic distance matrix. Statistical significance was assessed by permuting the rows and columns of the acoustic distance matrix 10,000 times and computing the fraction of permuted correlations exceeding the observed value.

PMI analysis Acoustic keywords were extracted from free-text acoustic descriptions by matching a fixed lexicon of 20 spectro-temporal terms (e.g., *tonal*, *broadband*, *high-frequency*). Semantic keywords were taken directly from the ontology keyword field. For each acoustic–semantic keyword pair (a, s), PMI was estimated as $\log_2 \hat{p}(a, s) - \log_2 \hat{p}(a) - \log_2 \hat{p}(s)$, where probabilities were estimated from call frequencies. Pairs with fewer than three co-occurrences were excluded.

Acoustic-to-semantic prediction We trained a ridge regression mapping acoustic embeddings $\mathbf{x} \in \mathbb{R}^{384}$ to semantic embeddings $\mathbf{y} \in \mathbb{R}^{384}$, regularized with ℓ_2 penalty λ selected by leave-one-species-out cross-validation. Predicted embeddings were ℓ_2-normalized before evaluation. Performance was measured by (i) mean cosine similarity between predicted and true semantic embeddings and (ii) mean reciprocal rank (MRR) of the true call's semantic embedding among all held-out calls, ranked by cosine similarity to the prediction. We additionally evaluated a shallow MLP (two hidden layers of 512 units, ReLU activations, dropout 0.2) and a retrieval baseline (k-nearest neighbors in acoustic space, semantic embedding averaged over the k neighbors). Cross-validation used 10 folds stratified at the species level.

Heavy AI methods This work made maximally heavy use of AI technology including database collection, curation and organization, discussion, design and implementation of analyses, literature search, writing and reviewing.

9 Contributions

Conceptualization: A.B., C.D. **Methodology:** A.B., C.D., E.F. **Software:** E.F., G.H. **Validation:** C.D., E.F. **Formal analysis:** A.B., E.F. **Investigation:** A.B., C.D. **Resources:** I.J. **Data curation:** A.B., C.D., E.F. **Writing – original draft:** A.B. **Writing – review & editing:** A.B., C.D., E.F., I.J. **Visualization:** E.F., G.H. **Supervision:** I.J. **Project administration:** I.J. **Funding acquisition:** I.J.

10 Acknowledgements

We thank the researchers whose detailed ethological and bioacoustic studies form the foundation of AnimalLex.

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A Web application

We release an interactive web application at https://emmanuel-dev.tail3ea00b.ts.net/ that allows users to: (i) browse AnimalLex filtered by species, taxon, semantic category, or acoustic keyword; (ii) translate any call to its nearest semantic neighbour in any other species; (iii) explore semantic and acoustic embedding spaces as interactive 3-D UMAP projections; and (iv) optionally include predicted-semantics calls from unannotated recordings. The application is built on [placeholder: framework] and is freely accessible without registration.

B Humans and the biological code

This appendix section applies AnimalLex's analytical framework to the human side of the animal kingdom in two complementary ways: first via abstract human-language glosses (anthropomorphic, exploratory), and second via a curated database of non-linguistic human vocal bursts that can be directly compared to animal calls on equal terms.

B.1 Human-language glosses (anthropomorphic, use with caution)

The translation framework can be applied with human language as one of the two "species": given a word or phrase, embed it in the shared semantic space and retrieve the animal call with the nearest embedding; conversely, given any call, retrieve the closest human-language gloss. This mapping is *not* a claim that animals share human concepts—it reflects only that the ethological descriptions in AnimalLex, written by human researchers, overlap in embedding space with human vocabulary. Every translation step through human language imposes an anthropomorphic frame and risks projecting a human *Umwelt* onto species whose perceptual and cognitive worlds may be radically different. We include Table 2 here as an exploratory tool for human audiences, not as a scientific claim.

Table 2. **Human-language translation: selected examples (see caveats above).** For each human phrase, the best-matching call in AnimalLex by semantic embedding similarity. [Placeholder: to be filled once translation pipeline is run.]

Human phrase	**Sim.**	**Best-match call**	**Acoustic description**
danger	—	[species] / [call]	[acoustic]
I am hungry	—	[species] / [call]	[acoustic]
come here	—	[species] / [call]	[acoustic]
I am here	—	[species] / [call]	[acoustic]
stay away	—	[species] / [call]	[acoustic]
greeting	—	[species] / [call]	[acoustic]
let's move together	—	[species] / [call]	[acoustic]
I am afraid	—	[species] / [call]	[acoustic]
this is mine	—	[species] / [call]	[acoustic]
help me	—	[species] / [call]	[acoustic]

B.2 Human non-linguistic vocal bursts

A stronger test of the biological code's universality is to include *human* non-linguistic vocalizations — short affective bursts such as laughter, gasps, sighs, growls, and cries — in the same framework as animal calls. Unlike language-mediated glosses, these vocalizations are pre-lexical, cross-culturally produced and recognized Sauter et al., 2010, and share evolutionary continuity with homologous calls in other primates. We curated a database of 30 human vocal burst types following the 30-category taxonomy of Cowen et al. (2019), who collected and rated 2,032 non-speech human vocal burst stimuli. Each burst type received the same annotation schema as animal calls in AnimalLex: an acoustic description (phonetic form, pitch contour, voice quality, duration) and a semantic description (affective context and social function), with ontology keywords drawn from the shared AnimalLex vocabulary (*alarm*, *affiliative*, *display*, *distress*, etc.) augmented by two new valence keywords (*positive_affect*, *negative_affect*).

Semantic correspondents. Figure 6 shows the maximum semantic cosine similarity between each of the 30 human burst types and each animal species in AnimalLex. Several interpretable patterns emerge. Fear and pain bursts — high-pitched, sudden, broadband exclamations — consistently map to alarm and anti-predator calls across all three vertebrate classes, with particularly high similarity to bat echolocation-adjacent distress calls and primate alarm barks. Contentment, serenity, and adoration bursts map most strongly to affiliative calls in social primates and songbirds, consistent with the predicted association between soft, low-frequency vocalizations and affiliation. The "Guilt" row, acoustically similar to a soft descending murmur, is weakest across the board, reflecting the absence of a clear non-human functional analogue.

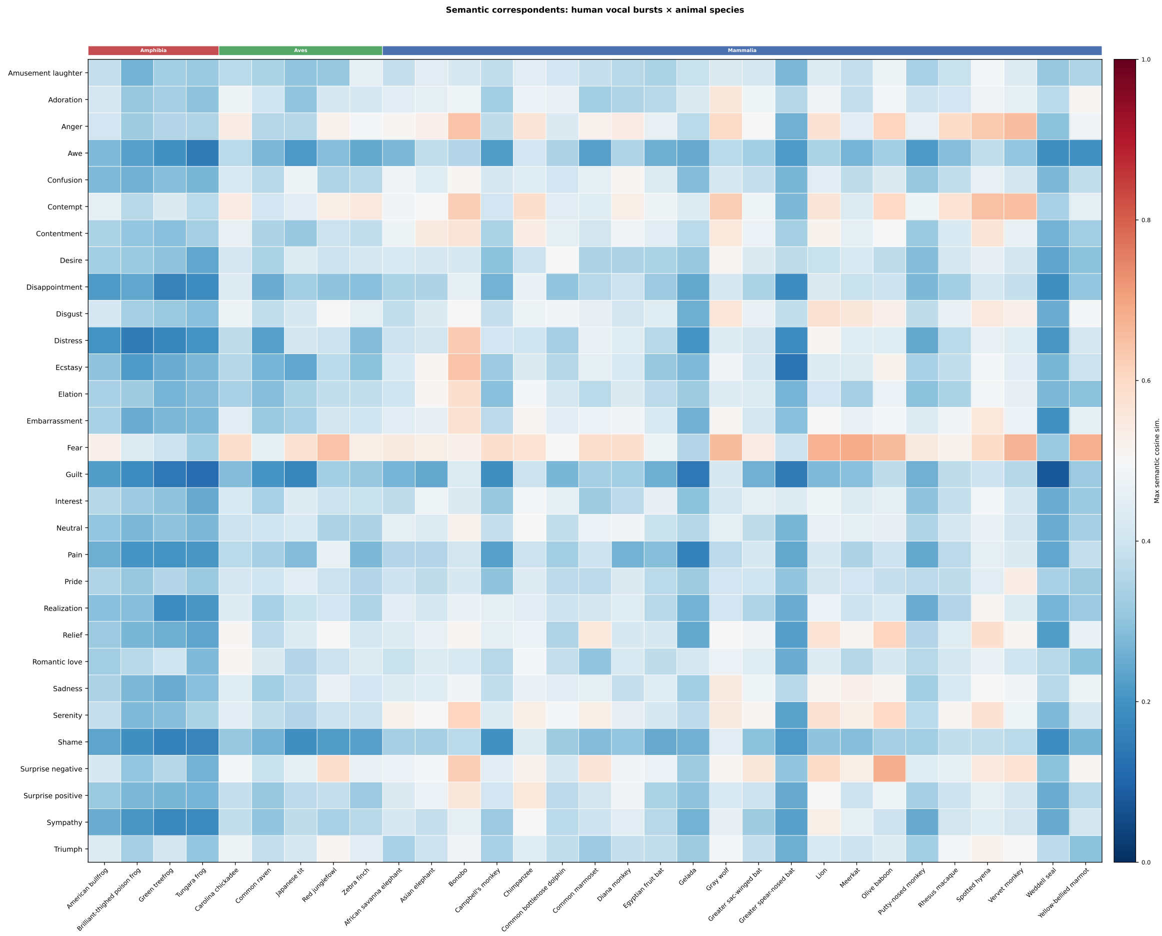

Figure 6. Semantic correspondents between human vocal bursts and animal species. Each cell shows the maximum semantic cosine similarity (L2-normalised sentence embeddings) between a human burst type (row) and any call of an animal species (column). Species are sorted by taxonomic class (Amphibia | Aves | Mammalia, colored bars on top) and alphabetically within class. Annotated cells (≥ 0.75) show the best-matching call name. Warm colors indicate close semantic correspondence.

Keyword-level form-to-meaning structure. Figure 7 applies the same PMI analysis as Fig. 4 to the human burst database. The acoustic rows are identical; the semantic columns reflect the human keyword vocabulary. Several patterns echo the biological code identified in non-human animals: *high-frequency* calls are strongly associated with *alarm* and *referential* functions (fear gasp, surprise burst, pain exclamation; PMI ≈ 2.3), paralleling the high-frequency $\times$ alarm enrichment across taxa. *Broadband/noisy* calls are enriched for *negative affect* (anger, disgust; PMI ≈ 1.3), consistent with Morton's frequency code Morton, 1977. *Repetitive* and *pulsed* bursts are enriched for *display* and *contact* functions (laughter; PMI ≈ 1.3), and *long/sustained* calls are depleted for *display* (soft sighs and murmurs are not attention-grabbing; PMI ≈ -1.7).

To quantify how much the overall pattern overlaps with the non-human animal PMI (Fig. 4), we computed the Pearson correlation between the two matrices restricted to their shared keywords (alarm, affiliative, display, contact, distress, referential): $\rho_{\mathrm{PMI}} = -0.03$. This *PMI pattern similarity* is a completely different quantity from the call-pair Mantel r reported elsewhere: it asks "do the same acoustic features get associated with the same semantic categories in both humans and animals?", not "does acoustic proximity predict semantic proximity across individual calls?" A near-zero ρ_{PMI} means that the *global rank-ordering* of acoustic–semantic weights differs: individual strong associations are preserved (high-frequency $\rightarrow$ alarm, broadband $\rightarrow$ negative affect), but their relative magnitudes and the allocation across categories diverge. The species-level analysis below reveals why: the code runs in opposite directions for mammals and birds, and their contributions cancel in the aggregate.

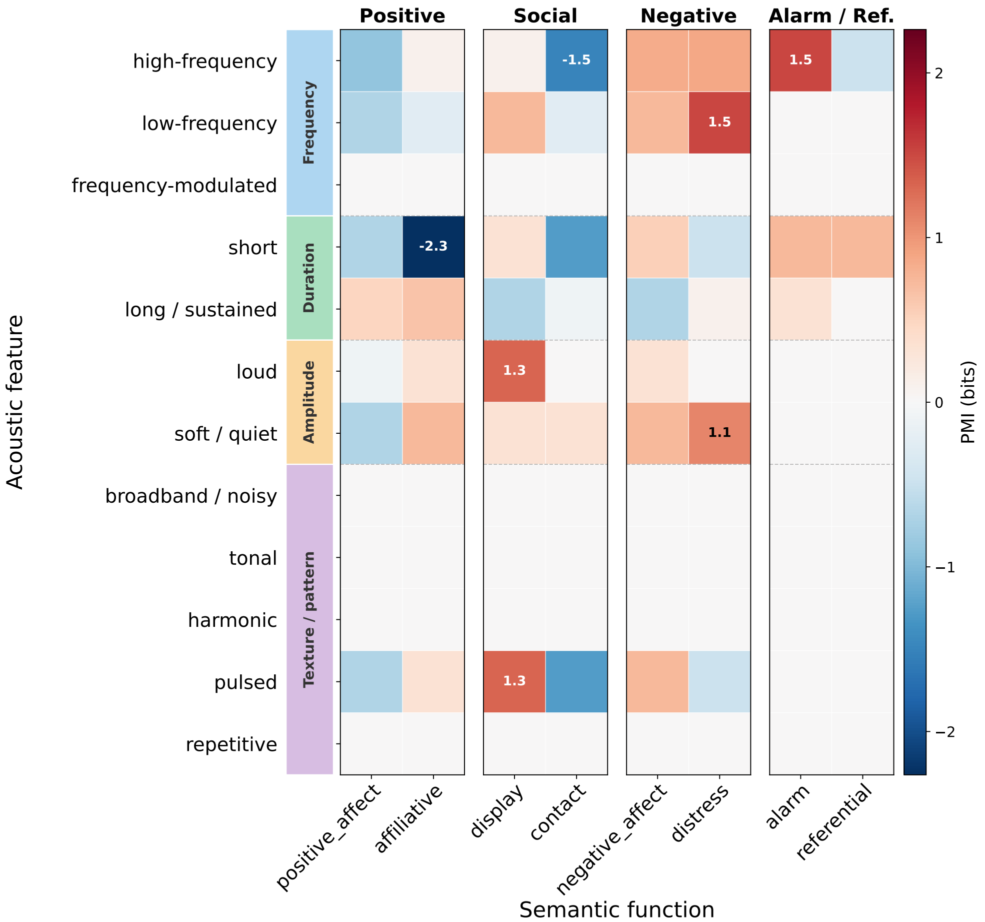

Figure 7. Keyword-level structure of the biological code in human vocal bursts. Same format as Fig. 4. Rows: 12 acoustic features grouped by type (colored strips). Columns: 8 semantic keywords surviving a minimum-count threshold ($\geq$ 3 calls), grouped into four panels. Numbers annotate the two strongest and one weakest association per panel.

Human vocalizations in the cross-species Mantel matrix. Figure 8 adds a "human row" to the species-pairwise Mantel matrix (Fig. 3): for each animal species, the Pearson r between acoustic and semantic cosine similarities across all $30 \times n_s$ human–animal call pairs. Within each class, species are sorted by their r value (highest first), revealing the graded structure of human-animal correspondence.

The two findings above are now reconcilable. *Mammals* show a consistently positive mean correlation ($\bar{r} = 0.082$), with the strongest alignment in a bat (Egyptian fruit bat, $r = 0.65$), a prosimian (Indri, $r = 0.53$), and several Old World monkeys and apes (Crowned guenon, Chacma baboon, Mandrill; $r \approx 0.38$–0.41). *Amphibians* are weakly positive ($\bar{r} = 0.039$). *Birds*, strikingly, are on average *negative* ($\bar{r} = -0.126$): for most avian species, the calls that sound most like human bursts tend to carry *opposite* semantic functions. This inversion — where the acoustic-semantic code runs in an opposing direction between birds and humans — is the main driver of the near-zero aggregate PMI correlation. Averaged across all three classes, their contributions partially cancel, bringing the keyword-level correlation to $r \approx 0$.

The picture that emerges is therefore: at the keyword level, the code appears globally flat because mammalian and avian influences offset each other; breaking by species exposes the underlying structure — genuine acoustic–semantic sharing with mammals (amplified by evolutionary and ecological proximity), and a largely inverted mapping with birds, whose vocal priorities (territory, mate attraction, species recognition via song) differ fundamentally from human affective burst functions.

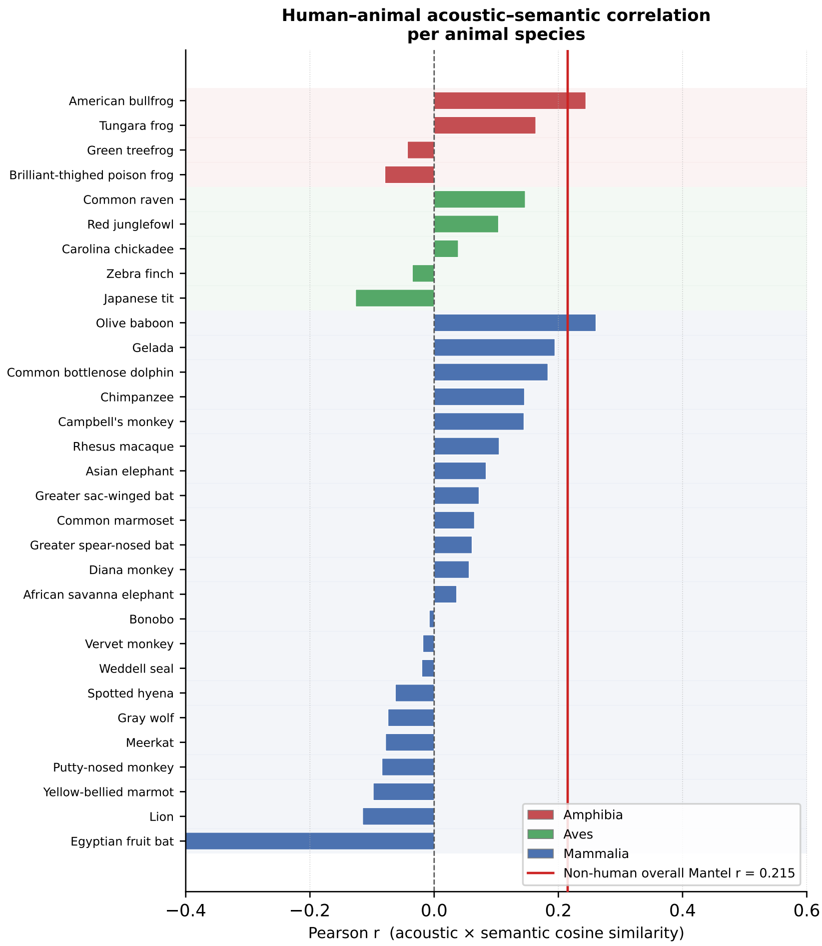

Figure 8. **Human vocal bursts in the cross-species Mantel matrix.** Each bar shows the Pearson r between acoustic and semantic cosine similarities for all human–animal call pairs involving that species ($30 \times n_s$ pairs; species with fewer than 6 pairs excluded). Within each taxonomic class, species are sorted by r value (highest first); class colors match Fig. 3. The red vertical line marks the non-human overall Mantel $r = 0.215$. Mammals are predominantly positive, amphibians weakly positive, and birds predominantly negative — their class-level averages partially cancel in aggregate keyword comparisons.

C Sensitivity analysis

Figure 9 presents two sensitivity analyses. Panel A shows Mantel r (with 95% bootstrap confidence intervals, $n_{\text{boot}} = 500$) computed separately for within-class and cross-class species pairs. The form-to-meaning alignment is positive and significant within Mammalia ($r = 0.364$), Aves ($r = 0.377$), and Amphibia ($r = 0.694$), and holds for most cross-class comparisons; the negative value for Aves $\times$ Amphibia reflects the small sample ($n = 731$ pairs).

Panel B quantifies how the mutual nearest-neighbour graph changes when the strictness of the cross-modal agreement criterion is relaxed from top-1 to top-k ($k \in \{1, 2, 3\}$). At $k = 1$ (the main analysis), 4 cliques span all three classes; relaxing to $k = 2$ yields 235 such cliques, and $k = 3$ yields 10,690. The main-text result is therefore conservative: the qualitative finding strengthens as the criterion is relaxed.

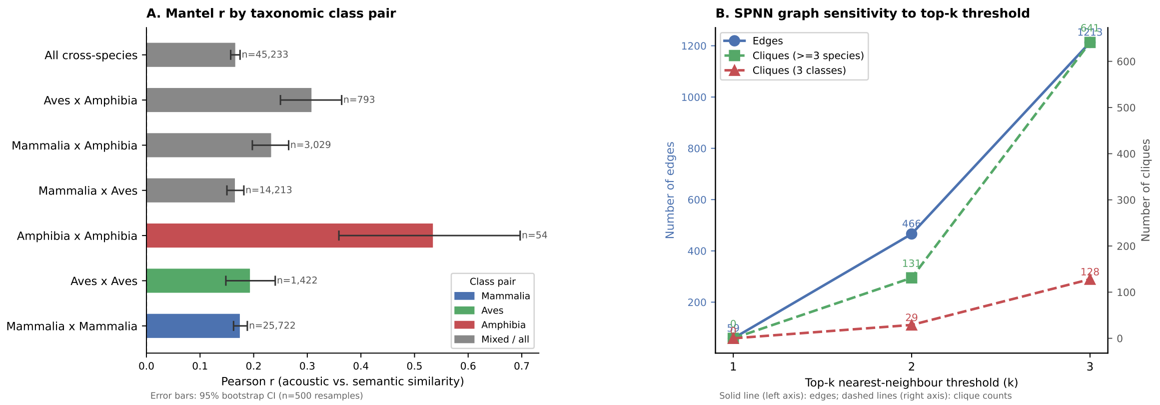

Figure 9. **Sensitivity analysis.** *Left*: Mantel r by taxonomic class pair. Bars show Pearson r; error bars are 95% bootstrap CIs. *Right*: Number of graph edges, cliques with ≥ 3 species, and three-class cliques as the mutual-NN threshold varies from top-1 (strict) to top-3 (relaxed).

D Bootstrap confidence intervals for Mantel statistics

Table 3 reproduces the Mantel test results from the main text, augmented with 95% bootstrap confidence intervals ($n_{\text{boot}} = 1{,}000$ for plain Mantel tests, $n_{\text{boot}} = 500$ for partial Mantel).

Table 3. Mantel test results with 95% bootstrap confidence intervals.

Test	r	95% CI	p
All pairs	+0.309	[+0.300, +0.319]	< 0.001
Within species (pooled)	+0.233	—	< 0.001
Same family, cross-species	+0.488	—	< 0.001
Cross-family	+0.233	—	< 0.001
Partial | species identity	+0.309	[+0.273, +0.388]	< 0.001
Partial | family identity	+0.296	[+0.261, +0.374]	< 0.001

E Validation against audio representations - IF WE CAN GET A SUBSET

[Placeholder: comparison of Mantel test results when acoustic embeddings are derived from (a) text descriptions, as in the main analysis, vs. (b) audio features extracted by AVES Hagiwara, 2022 or NatureLM-audio Robinson et al., 2025 for the subset of calls for which verified audio samples are available. This analysis tests whether the form-to-meaning alignment detected through textual descriptions is an artifact of description style or reflects genuine acoustic structure.]

F More database statistics

See Figure 10

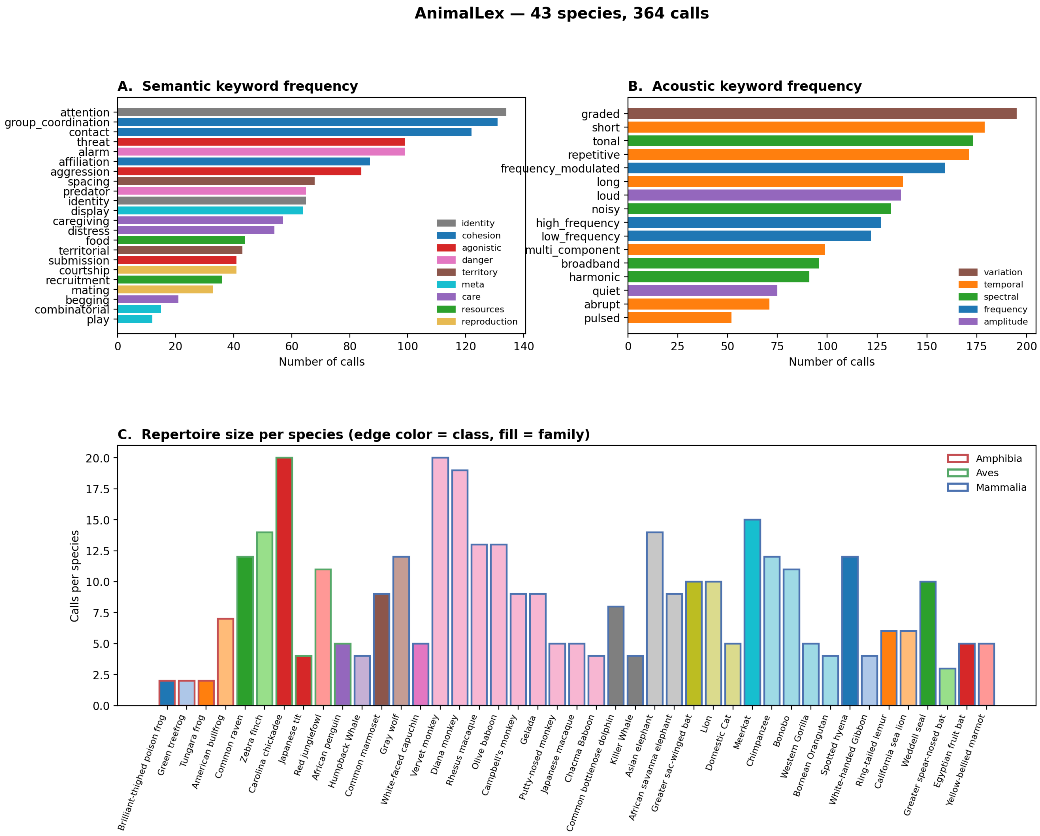

Figure 10. **Calls per species in AnimalLex**, sorted by count. Color encodes taxonomic class. The median repertoire size is [X] calls; the largest repertoire (*Loxodonta africana*) contains [X] calls.

G Complete cross-species call groups

Figure 11 summarises all 297 cross-species call groups (maximal cliques in the mutual-NN graph with ≥ 2 distinct species) identified in AnimalLex. Groups range in size from 2 to 26 calls, and span all five semantic categories. Cohesion groups are the most frequent (134), followed by social (60), danger (49), and distress (44).

Table 4 lists all groups sorted by taxonomic diversity (number of classes, then number of species).

Table 4. All cross-species call groups (maximal cliques in the mutual nearest-neighbour graph with ≥ 2 distinct species). Sorted by number of taxonomic classes, then families, then group size. Icons denote taxonomic class (🐸 Amphibia, 🐦 Aves, 🐾 Mammalia).

Classes	Families	Species	Category	Members
3	3	7	cohesion	🐸 *Engystomops pustulosus* (Whine–chuck advertisement call) 🐦 *Pheugopedius euophrys* (Contact call) 🐾 *Cercopithecus pogonias* (Grunt) 🐾 *Cercopithecus neglectus* (Grunt) 🐾 *Cercopithecus mona* (Grunt) 🐾 *Erythrocebus patas* (Grunt) 🐾 *Miopithecus talapoin* (Grunt)
3	3	3	social	🐸 *Dendrobatidae; various genera* (Aggressive call) 🐦 *Sturnus vulgaris* (Alarm call) 🐾 *Gorilla beringei* (Scream)

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Classes	Families	Species	Category	Members
3	3	3	cohesion	*Engystomops pustulosus* (Whine–chuck advertisement call) *Pheugopedius euophrys* (Contact call) *Cercopithecus ascanius* (Boom)
3	3	3	social	*Lithobates catesbeianus* (Advertisement call) *Pheugopedius euophrys* (Coordinated duet song) *Symphalangus syndactylus* (Great call (female-led duet))
2	8	9	foraging	*Psittacus erithacus* (Begging call) *Lonchura striata domestica* (Begging call) *Poecile carolinensis* (Begging call) *Pomatostomus ruficeps* (Begging call) *Sturnus vulgaris* (Begging call) *Plocepasser mahali* (Begging call) *Taeniopygia guttata* (Begging call) *Saccopteryx bilineata* (Mother–infant isolation call) *Suricata suricatta* (Infant begging call)
2	7	7	foraging	*Psittacus erithacus* (Begging call) *Poecile carolinensis* (Begging call) *Pomatostomus ruficeps* (Begging call) *Sturnus vulgaris* (Begging call) *Plocepasser mahali* (Begging call) *Saccopteryx bilineata* (Mother–infant isolation call) *Alouatta spp.* (Infant distress call)
2	7	7	foraging	*Poecile carolinensis* (Begging call) *Pomatostomus ruficeps* (Begging call) *Sturnus vulgaris* (Begging call) *Plocepasser mahali* (Begging call) *Eulemur spp.* (Infant distress call) *Suricata suricatta* (Infant begging call) *Cercopithecus ascanius* (Infant distress call)
2	7	7	danger	*Pomatostomus ruficeps* (Alarm call) *Sturnus vulgaris* (Alarm call) *Plocepasser mahali* (Alarm call) *Papio ursinus* (Bark) *Callithrix jacchus* (Tsik (alarm call)) *Lemur catta* (Alarm call (general)) *Sapajus apella* (Alarm call)
2	6	7	foraging	*Poecile carolinensis* (Begging call) *Pomatostomus ruficeps* (Begging call) *Sturnus vulgaris* (Begging call) *Plocepasser mahali* (Begging call) *Eulemur spp.* (Infant distress call) *Cercopithecus diana* (Infant distress call) *Cercopithecus ascanius* (Infant distress call)
2	6	7	danger	*Pomatostomus ruficeps* (Alarm call) *Sturnus vulgaris* (Alarm call) *Plocepasser mahali* (Alarm call) *Eulemur spp.* (Alarm call) *Papio ursinus* (Bark) *Lemur catta* (Alarm call (general)) *Sapajus apella* (Alarm call)

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Classes	Families	Species	Category	Members
2	6	6	danger	*Pomatostomus ruficeps* (Alarm call) *Sturnus vulgaris* (Alarm call) *Plocepasser mahali* (Alarm call) *Callithrix jacchus* (Tsik (alarm call)) *Microcebus spp.* (Alarm call) *Lemur catta* (Alarm call (general))
2	5	13	distress	*Psittacus erithacus* (Distress scream) *Pan paniscus* (Scream) *Callithrix jacchus* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Theropithecus gelada* (Scream) *Alouatta spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Mandrillus sphinx* (Scream) *Cercopithecus mona* (Scream) *Cercopithecus nictitans* (Scream) *Cercopithecus ascanius* (Scream) *Miopithecus talapoin* (Scream)
2	5	8	distress	*Poecile carolinensis* (Begging call) *Sturnus vulgaris* (Begging call) *Eulemur spp.* (Infant distress call) *Mandrillus leucophaeus* (Infant distress call) *Mandrillus sphinx* (Infant distress call) *Suricata suricatta* (Infant begging call) *Erythrocebus patas* (Infant distress call) *Cercopithecus ascanius* (Infant distress call)
2	5	7	distress	*Lithobates catesbeianus* (Distress scream) *Hyla cinerea* (Distress call) *Eulemur spp.* (Scream) *Mandrillus leucophaeus* (Scream) *Microcebus spp.* (Scream) *Erythrocebus patas* (Scream) *Piliocolobus spp.* (Scream)
2	5	7	danger	*Sturnus vulgaris* (Alarm call) *Plocepasser mahali* (Alarm call) *Eulemur spp.* (Alarm call) *Papio ursinus* (Bark) *Papio anubis* (Bark) *Lemur catta* (Alarm call (general)) *Sapajus apella* (Alarm call)
2	5	6	cohesion	*Psittacus erithacus* (Contact call) *Poecile carolinensis* (Contact call) *Parus minor* (Contact call) *Pheugopedius euophrys* (Contact call) *Plocepasser mahali* (Contact call) *Saccopteryx bilineata* (Contact call)
2	5	6	foraging	*Poecile carolinensis* (Begging call) *Sturnus vulgaris* (Begging call) *Taeniopygia guttata* (Begging call) *Mandrillus leucophaeus* (Infant distress call) *Mandrillus sphinx* (Infant distress call) *Suricata suricatta* (Infant begging call)

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Classes	Families	Species	Category	Members
2	5	6	foraging	*Pomatostomus ruficeps* (Begging call) *Plocepasser mahali* (Begging call) *Eulemur spp.* (Infant distress call) *Cercopithecus pogonias* (Infant distress call) *Suricata suricatta* (Infant begging call) *Cercopithecus ascanius* (Infant distress call)
2	5	6	danger	*Pomatostomus ruficeps* (Alarm call) *Sturnus vulgaris* (Alarm call) *Plocepasser mahali* (Alarm call) *Eulemur spp.* (Alarm call) *Microcebus spp.* (Alarm call) *Lemur catta* (Alarm call (general))
2	5	5	foraging	*Poecile carolinensis* (Begging call) *Sturnus vulgaris* (Begging call) *Saccopteryx bilineata* (Mother–infant isolation call) *Suricata suricatta* (Infant begging call) *Crocuta crocuta* (Cub distress call)
2	4	17	distress	*Psittacus erithacus* (Distress scream) *Colobus guereza* (Scream) *Papio ursinus* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Cercopithecus diana* (Scream) *Theropithecus gelada* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Mandrillus sphinx* (Scream) *Cercopithecus mona* (Scream) *Cercopithecus nictitans* (Scream) *Cercopithecus ascanius* (Scream) *Lemur catta* (Scream) *Sapajus apella* (Scream) *Chlorocebus pygerythrus* (Scream) *Papio cynocephalus* (Scream)
2	4	17	distress	*Psittacus erithacus* (Distress scream) *Colobus guereza* (Scream) *Papio ursinus* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Cercopithecus diana* (Scream) *Theropithecus gelada* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Mandrillus sphinx* (Scream) *Cercopithecus mona* (Scream) *Cercopithecus nictitans* (Scream) *Cercopithecus ascanius* (Scream) *Lemur catta* (Scream) *Miopithecus talapoin* (Scream) *Sapajus apella* (Scream) *Chlorocebus pygerythrus* (Scream)

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Classes	Families	Species	Category	Members
2	4	17	distress	*Psittacus erithacus* (Distress scream) *Colobus guereza* (Scream) *Papio ursinus* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Cercopithecus diana* (Scream) *Theropithecus gelada* (Scream) *Alouatta spp.* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Mandrillus sphinx* (Scream) *Cercopithecus mona* (Scream) *Cercopithecus nictitans* (Scream) *Cercopithecus ascanius* (Scream) *Lemur catta* (Scream) *Miopithecus talapoin* (Scream) *Chlorocebus pygerythrus* (Scream)
2	4	15	distress	*Psittacus erithacus* (Distress scream) *Colobus guereza* (Scream) *Papio ursinus* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Theropithecus gelada* (Scream) *Alouatta spp.* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Mandrillus sphinx* (Scream) *Cercopithecus mona* (Scream) *Cercopithecus ascanius* (Scream) *Miopithecus talapoin* (Scream) *Chlorocebus pygerythrus* (Scream) *Gorilla gorilla* (Scream (loud scream))
2	4	15	distress	*Psittacus erithacus* (Distress scream) *Colobus guereza* (Scream) *Papio ursinus* (Scream) *Callithrix jacchus* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Theropithecus gelada* (Scream) *Alouatta spp.* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Mandrillus sphinx* (Scream) *Cercopithecus mona* (Scream) *Cercopithecus nictitans* (Scream) *Cercopithecus ascanius* (Scream) *Miopithecus talapoin* (Scream)

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Classes	Families	Species	Category	Members
2	4	15	distress	🐸 *Hyla cinerea* (Distress call) 🐾 *Colobus guereza* (Scream) 🐾 *Eulemur spp.* (Scream) 🐾 *Cercopithecus pogonias* (Scream) 🐾 *Cercopithecus neglectus* (Scream) 🐾 *Mandrillus leucophaeus* (Scream) 🐾 *Semnopithecus spp. / Trachypithecus spp.* (Scream) 🐾 *Allochrocebus lhoesti* (Scream) 🐾 *Erythrocebus patas* (Scream) 🐾 *Cercopithecus nictitans* (Scream) 🐾 *Piliocolobus spp.* (Scream) 🐾 *Cercopithecus ascanius* (Scream) 🐾 *Lemur catta* (Scream) 🐾 *Sapajus apella* (Scream) 🐾 *Chlorocebus pygerythrus* (Scream)
2	4	15	distress	🐸 *Hyla cinerea* (Distress call) 🐾 *Colobus guereza* (Scream) 🐾 *Eulemur spp.* (Scream) 🐾 *Cercopithecus pogonias* (Scream) 🐾 *Cercopithecus neglectus* (Scream) 🐾 *Mandrillus leucophaeus* (Scream) 🐾 *Alouatta spp.* (Scream) 🐾 *Semnopithecus spp. / Trachypithecus spp.* (Scream) 🐾 *Allochrocebus lhoesti* (Scream) 🐾 *Erythrocebus patas* (Scream) 🐾 *Cercopithecus nictitans* (Scream) 🐾 *Piliocolobus spp.* (Scream) 🐾 *Cercopithecus ascanius* (Scream) 🐾 *Lemur catta* (Scream) 🐾 *Chlorocebus pygerythrus* (Scream)
2	4	13	distress	🐸 *Hyla cinerea* (Distress call) 🐾 *Colobus guereza* (Scream) 🐾 *Cercopithecus pogonias* (Scream) 🐾 *Cercopithecus neglectus* (Scream) 🐾 *Mandrillus leucophaeus* (Scream) 🐾 *Alouatta spp.* (Scream) 🐾 *Semnopithecus spp. / Trachypithecus spp.* (Scream) 🐾 *Allochrocebus lhoesti* (Scream) 🐾 *Erythrocebus patas* (Scream) 🐾 *Piliocolobus spp.* (Scream) 🐾 *Cercopithecus ascanius* (Scream) 🐾 *Chlorocebus pygerythrus* (Scream) 🐾 *Gorilla gorilla* (Scream (loud scream))
2	4	9	distress	🐦 *Poecile carolinensis* (Begging call) 🐦 *Sturnus vulgaris* (Begging call) 🐾 *Eulemur spp.* (Infant distress call) 🐾 *Cercopithecus diana* (Infant distress call) 🐾 *Mandrillus leucophaeus* (Infant distress call) 🐾 *Mandrillus sphinx* (Infant distress call) 🐾 *Erythrocebus patas* (Infant distress call) 🐾 *Cercopithecus ascanius* (Infant distress call) 🐾 *Lemur catta* (Infant distress call)

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Classes	Families	Species	Category	Members
2	4	8	distress	*Hyla cinerea* (Distress call) *Colobus guereza* (Scream) *Eulemur spp.* (Scream) *Mandrillus leucophaeus* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Microcebus spp.* (Scream) *Erythrocebus patas* (Scream) *Piliocolobus spp.* (Scream)
2	4	7	danger	*Sturnus vulgaris* (Alarm call) *Plocepasser mahali* (Alarm call) *Eulemur spp.* (Alarm call) *Papio ursinus* (Bark) *Theropithecus gelada* (Alarm bark) *Papio anubis* (Bark) *Lemur catta* (Alarm call (general))
2	4	7	danger	*Sturnus vulgaris* (Alarm call) *Plocepasser mahali* (Alarm call) *Eulemur spp.* (Alarm call) *Papio ursinus* (Bark) *Theropithecus gelada* (Alarm bark) *Cercopithecus mona* (Alarm bark) *Lemur catta* (Alarm call (general))
2	4	7	distress	*Pomatostomus ruficeps* (Begging call) *Plocepasser mahali* (Begging call) *Eulemur spp.* (Infant distress call) *Cercopithecus pogonias* (Infant distress call) *Cercopithecus diana* (Infant distress call) *Cercopithecus nictitans* (Infant distress call) *Cercopithecus ascanius* (Infant distress call)
2	4	7	distress	*Psittacus erithacus* (Distress scream) *Lonchura striata domestica* (Distress call) *Colobus guereza* (Scream) *Theropithecus gelada* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Cercopithecus ascanius* (Scream) *Gorilla gorilla* (Scream (loud scream))
2	4	7	distress	*Psittacus erithacus* (Distress scream) *Lonchura striata domestica* (Distress call) *Taeniopygia guttata* (Distress call) *Colobus guereza* (Scream) *Theropithecus gelada* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Gorilla gorilla* (Scream (loud scream))
2	4	6	cohesion	*Lonchura striata domestica* (Tet call) *Poecile carolinensis* (Contact call) *Parus minor* (Contact call) *Taeniopygia guttata* (Tet call) *Saccopteryx bilineata* (Contact call) *Suricata suricatta* (Close-range contact call)

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Classes	Families	Species	Category	Members
2	4	6	danger	*Pomatostomus ruficeps* (Alarm call) *Sturnus vulgaris* (Alarm call) *Plocepasser mahali* (Alarm call) *Papio ursinus* (Bark) *Macaca fuscata* (Bark) *Macaca mulatta* (Bark)
2	4	5	cohesion	*Psittacus erithacus* (Contact call) *Poecile carolinensis* (Contact call) *Parus minor* (Contact call) *Saccopteryx bilineata* (Contact call) *Suricata suricatta* (Close-range contact call)
2	4	5	cohesion	*Lonchura striata domestica* (Tet call) *Poecile carolinensis* (Contact call) *Parus minor* (Contact call) *Plocepasser mahali* (Contact call) *Saccopteryx bilineata* (Contact call)
2	4	5	social	*Psittacus erithacus* (Alarm call) *Alouatta spp.* (Wrah / Growl) *Macaca fuscata* (Threat growl) *Macaca mulatta* (Threat growl) *Crocuta crocuta* (Growl)
2	4	5	social	*Engystomops pustulosus* (Aggressive call) *Alouatta spp.* (Wrah / Growl) *Macaca fuscata* (Threat growl) *Macaca mulatta* (Threat growl) *Crocuta crocuta* (Growl)
2	4	5	cohesion	*Lonchura striata domestica* (Tet call) *Parus minor* (Contact call) *Taeniopygia guttata* (Tet call) *Saccopteryx bilineata* (Contact call) *Cercopithecus nictitans* (Contact call (soft contact call))
2	4	5	cohesion	*Lonchura striata domestica* (Tet call) *Sturnus vulgaris* (Contact call) *Plocepasser mahali* (Contact call) *Colobus guereza* (Grunt) *Papio cynocephalus* (Grunt)
2	4	5	cohesion	*Sturnus vulgaris* (Contact call) *Pheugopedius euophrys* (Contact call) *Plocepasser mahali* (Contact call) *Colobus guereza* (Grunt) *Papio cynocephalus* (Grunt)
2	4	5	cohesion	*Plocepasser mahali* (Contact call) *Eulemur spp.* (Contact call) *Macaca fuscata* (Coo) *Macaca mulatta* (Coo) *Sapajus apella* (Contact call (chirp / peep))
2	4	5	cohesion	*Sturnus vulgaris* (Contact call) *Plocepasser mahali* (Contact call) *Macaca fuscata* (Coo) *Macaca mulatta* (Coo) *Sapajus apella* (Contact call (chirp / peep))

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Classes	Families	Species	Category	Members
2	4	4	cohesion	*Parus minor* (Contact call) *Saccopteryx bilineata* (Contact call) *Suricata suricatta* (Close-range contact call) *Cercopithecus mona* (Boom)
2	4	4	social	*Engystomops pustulosus* (Aggressive call) *Alouatta spp.* (Wrah / Growl) *Microcebus spp.* (Agonistic call) *Crocuta crocuta* (Growl)
2	4	4	social	*Lithobates catesbeianus* (Advertisement call) *Dendrobatidae; various genera* (Advertisement call) *Cercopithecus neglectus* (Boom) *Hylobates spp., Nomascus spp., Hoolock spp.* (Male solo song)
2	4	4	cohesion	*Lonchura striata domestica* (Tet call) *Parus minor* (Contact call) *Rousettus aegyptiacus* (Affiliative contact call) *Saccopteryx bilineata* (Contact call)
2	4	4	cohesion	*Plocepasser mahali* (Contact call) *Eulemur spp.* (Contact call) *Saccopteryx bilineata* (Contact call) *Macaca fuscata* (Coo)
2	4	4	cohesion	*Parus minor* (Contact call) *Plocepasser mahali* (Contact call) *Saccopteryx bilineata* (Contact call) *Macaca fuscata* (Coo)
2	4	4	cohesion	*Parus minor* (Contact call) *Pheugopedius euophrys* (Contact call) *Saccopteryx bilineata* (Contact call) *Cercopithecus nictitans* (Contact call (soft contact call))
2	4	4	cohesion	*Parus minor* (Contact call) *Saccopteryx bilineata* (Contact call) *Cercopithecus mona* (Boom) *Microcebus spp.* (Audible contact call)
2	4	4	cohesion	*Parus minor* (Contact call) *Pheugopedius euophrys* (Contact call) *Plocepasser mahali* (Contact call) *Cercopithecus ascanius* (Boom)
2	4	4	cohesion	*Sturnus vulgaris* (Contact call) *Plocepasser mahali* (Contact call) *Lemur catta* (Contact call (grunt / short contact call)) *Sapajus apella* (Contact call (chirp / peep))
2	4	4	cohesion	*Sturnus vulgaris* (Contact call) *Plocepasser mahali* (Contact call) *Colobus guereza* (Grunt) *Sapajus apella* (Contact call (chirp / peep))
2	4	4	cohesion	*Sturnus vulgaris* (Contact call) *Plocepasser mahali* (Contact call) *Lemur catta* (Contact call (grunt / short contact call)) *Papio cynocephalus* (Grunt)
2	4	4	danger	*Pomatostomus ruficeps* (Alarm call) *Plocepasser mahali* (Alarm call) *Eulemur spp.* (Alarm call) *Semnopithecus spp. / Trachypithecus spp.* (Alarm call)

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Classes	Families	Species	Category	Members
2	4	4	danger	*Pomatostomus ruficeps* (Alarm call) *Sturnus vulgaris* (Alarm call) *Plocepasser mahali* (Alarm call) *Gorilla beringei* (Scream)
2	3	11	cohesion	*Pheugopedius euophrys* (Contact call) *Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Gorilla beringei* (Grunt (close call class)) *Macaca fuscata* (Grunt) *Allochrocebus lhoesti* (Grunt) *Cercopithecus mona* (Grunt) *Erythrocebus patas* (Grunt) *Macaca mulatta* (Grunt) *Miopithecus talapoin* (Grunt)
2	3	10	cohesion	*Pheugopedius euophrys* (Contact call) *Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Gorilla beringei* (Grunt (close call class)) *Macaca fuscata* (Grunt) *Allochrocebus lhoesti* (Grunt) *Cercopithecus mona* (Grunt) *Erythrocebus patas* (Grunt) *Macaca mulatta* (Grunt) *Papio cynocephalus* (Grunt)
2	3	10	cohesion	*Pheugopedius euophrys* (Contact call) *Colobus guereza* (Grunt) *Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Gorilla beringei* (Grunt (close call class)) *Semnopithecus spp. / Trachypithecus spp.* (Grunt) *Allochrocebus lhoesti* (Grunt) *Piliocolobus spp.* (Grunt) *Miopithecus talapoin* (Grunt) *Chlorocebus pygerythrus* (Grunt)
2	3	9	cohesion	*Pheugopedius euophrys* (Contact call) *Colobus guereza* (Grunt) *Cercopithecus neglectus* (Grunt) *Gorilla beringei* (Grunt (close call class)) *Semnopithecus spp. / Trachypithecus spp.* (Grunt) *Allochrocebus lhoesti* (Grunt) *Piliocolobus spp.* (Grunt) *Chlorocebus pygerythrus* (Grunt) *Papio cynocephalus* (Grunt)
2	3	9	cohesion	*Pheugopedius euophrys* (Contact call) *Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Gorilla beringei* (Grunt (close call class)) *Allochrocebus lhoesti* (Grunt) *Erythrocebus patas* (Grunt) *Miopithecus talapoin* (Grunt) *Chlorocebus pygerythrus* (Grunt)

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Classes	Families	Species	Category	Members
2	3	8	cohesion	*Pheugopedius euophrys* (Contact call) *Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Gorilla beringei* (Grunt (close call class)) *Allochrocebus lhoesti* (Grunt) *Erythrocebus patas* (Grunt) *Chlorocebus pygerythrus* (Grunt) *Papio cynocephalus* (Grunt)
2	3	7	cohesion	*Parus minor* (Contact call) *Mandrillus leucophaeus* (Male loud call) *Mandrillus sphinx* (Male loud call) *Suricata suricatta* (Close-range contact call) *Cercopithecus mona* (Boom) *Papio anubis* (Male loud call) *Erythrocebus patas* (Male loud call)
2	3	7	distress	*Lonchura striata domestica* (Distress call) *Colobus guereza* (Scream) *Theropithecus gelada* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Piliocolobus spp.* (Scream) *Cercopithecus ascanius* (Scream) *Gorilla gorilla* (Scream (loud scream))
2	3	7	distress	*Lonchura striata domestica* (Distress call) *Taeniopygia guttata* (Distress call) *Colobus guereza* (Scream) *Theropithecus gelada* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Piliocolobus spp.* (Scream) *Gorilla gorilla* (Scream (loud scream))
2	3	6	cohesion	*Lonchura striata domestica* (Tet call) *Plocepasser mahali* (Contact call) *Colobus guereza* (Grunt) *Semnopithecus spp. / Trachypithecus spp.* (Grunt) *Piliocolobus spp.* (Grunt) *Papio cynocephalus* (Grunt)
2	3	6	cohesion	*Parus minor* (Contact call) *Cercopithecus neglectus* (Boom) *Allochrocebus lhoesti* (Boom) *Cercopithecus mona* (Boom) *Microcebus spp.* (Audible contact call) *Cercopithecus ascanius* (Boom)
2	3	6	cohesion	*Pheugopedius euophrys* (Contact call) *Plocepasser mahali* (Contact call) *Colobus guereza* (Grunt) *Semnopithecus spp. / Trachypithecus spp.* (Grunt) *Piliocolobus spp.* (Grunt) *Papio cynocephalus* (Grunt)
2	3	6	danger	*Sturnus vulgaris* (Alarm call) *Plocepasser mahali* (Alarm call) *Papio ursinus* (Bark) *Macaca fuscata* (Bark) *Cercopithecus mona* (Alarm bark) *Macaca mulatta* (Bark)

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Classes	Families	Species	Category	Members
2	3	5	cohesion	*Plocepasser mahali* (Contact call) *Colobus guereza* (Grunt) *Semnopithecus spp. / Trachypithecus spp.* (Grunt) *Piliocolobus spp.* (Grunt) *Sapajus apella* (Contact call (chirp / peep))
2	3	5	danger	*Pomatostomus ruficeps* (Alarm call) *Plocepasser mahali* (Alarm call) *Macaca fuscata* (Bark) *Semnopithecus spp. / Trachypithecus spp.* (Alarm call) *Macaca mulatta* (Bark)
2	3	5	danger	*Plocepasser mahali* (Alarm call) *Eulemur spp.* (Alarm call) *Theropithecus gelada* (Alarm bark) *Semnopithecus spp. / Trachypithecus spp.* (Alarm call) *Cercopithecus mona* (Alarm bark)
2	3	5	danger	*Plocepasser mahali* (Alarm call) *Papio ursinus* (Bark) *Cercopithecus pogonias* (Hack (sharp warning/alarm call)) *Papio anubis* (Bark) *Lemur catta* (Alarm call (general))
2	3	4	cohesion	*Parus minor* (Contact call) *Suricata suricatta* (Close-range contact call) *Cercopithecus mona* (Boom) *Cercopithecus ascanius* (Boom)
2	3	4	cohesion	*Lonchura striata domestica* (Tet call) *Taeniopygia guttata* (Tet call) *Papio ursinus* (Male loud call) *Suricata suricatta* (Close-range contact call)
2	3	4	social	*Lithobates catesbeianus* (Advertisement call) *Dendrobatidae; various genera* (Advertisement call) *Colobus guereza* (Roar) *Cercopithecus neglectus* (Boom)
2	3	4	social	*Engystomops pustulosus* (Aggressive call) *Rousettus aegyptiacus* (Social squabble call) *Macaca fuscata* (Threat growl) *Macaca mulatta* (Threat growl)
2	3	4	cohesion	*Lithobates catesbeianus* (Advertisement call) *Cercopithecus neglectus* (Boom) *Hylobates spp., Nomascus spp., Hoolock spp.* (Male solo song) *Allochrocebus lhoesti* (Boom)
2	3	4	social	*Psittacus erithacus* (Alarm call) *Rousettus aegyptiacus* (Social squabble call) *Macaca fuscata* (Threat growl) *Macaca mulatta* (Threat growl)
2	3	4	cohesion	*Lonchura striata domestica* (Tet call) *Mandrillus leucophaeus* (Grunt) *Rousettus aegyptiacus* (Affiliative contact call) *Mandrillus sphinx* (Grunt)
2	3	4	cohesion	*Parus minor* (Contact call) *Saccopteryx bilineata* (Contact call) *Cercopithecus mona* (Boom) *Cercopithecus nictitans* (Contact call (soft contact call))

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Classes	Families	Species	Category	Members
2	3	4	cohesion	*Lonchura striata domestica* (Tet call) *Sturnus vulgaris* (Contact call) *Colobus guereza* (Grunt) *Cercopithecus diana* (Contact call (soft contact call))
2	3	4	cohesion	*Parus minor* (Contact call) *Plocepasser mahali* (Contact call) *Macaca fuscata* (Coo) *Macaca mulatta* (Coo)
2	3	4	cohesion	*Sturnus vulgaris* (Contact call) *Pheugopedius euophrys* (Contact call) *Colobus guereza* (Grunt) *Cercopithecus diana* (Contact call (soft contact call))
2	3	3	cohesion	*Poecile carolinensis* (Contact call) *Plocepasser mahali* (Contact call) *Lemur catta* (Contact call (grunt / short contact call))
2	3	3	social	*Lithobates catesbeianus* (Encounter aggression call) *Microcebus spp.* (Agonistic call) *Crocuta crocuta* (Growl)
2	3	3	social	*Lithobates catesbeianus* (Encounter aggression call) *Canis lupus* (Growl) *Crocuta crocuta* (Growl)
2	3	3	social	*Engystomops pustulosus* (Aggressive call) *Canis lupus* (Growl) *Crocuta crocuta* (Growl)
2	3	3	cohesion	*Dendrobatidae; various genera* (Advertisement call) *Cercopithecus neglectus* (Boom) *Microcebus spp.* (Audible contact call)
2	3	3	social	*Lithobates catesbeianus* (Advertisement call) *Dendrobatidae; various genera* (Advertisement call) *Pheugopedius euophrys* (Coordinated duet song)
2	3	3	social	*Dendrobatidae; various genera* (Aggressive call) *Engystomops pustulosus* (Aggressive call) *Gorilla beringei* (Scream)
2	3	3	social	*Engystomops pustulosus* (Aggressive call) *Rousettus aegyptiacus* (Social squabble call) *Canis lupus* (Growl)
2	3	3	social	*Engystomops pustulosus* (Aggressive call) *Gorilla beringei* (Scream) *Microcebus spp.* (Agonistic call)
2	3	3	social	*Lithobates catesbeianus* (Encounter aggression call) *Saccopteryx bilineata* (Aggressive call) *Microcebus spp.* (Agonistic call)
2	3	3	social	*Lithobates catesbeianus* (Encounter aggression call) *Gorilla beringei* (Scream) *Microcebus spp.* (Agonistic call)
2	3	3	social	*Lithobates catesbeianus* (Encounter aggression call) *Rousettus aegyptiacus* (Social squabble call) *Saccopteryx bilineata* (Aggressive call)
2	3	3	social	*Lithobates catesbeianus* (Encounter aggression call) *Rousettus aegyptiacus* (Social squabble call) *Canis lupus* (Growl)

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Classes	Families	Species	Category	Members
2	3	3	distress	*Lithobates catesbeianus* (Distress scream) *Hyla cinerea* (Distress call) *Hylobates spp., Nomascus spp., Hoolock spp.* (Alarm scream)
2	3	3	danger	*Pomatostomus ruficeps* (Alarm call) *Canis lupus* (Bark) *Lemur catta* (Alarm call (general))
2	3	3	cohesion	*Psittacus erithacus* (Contact call) *Parus minor* (Contact call) *Gorilla gorilla* (Hoot series (long-distance call))
2	3	3	cohesion	*Pheugopedius euophrys* (Contact call) *Pan troglodytes* (Pant-hoot) *Cercopithecus ascanius* (Boom)
2	3	3	cohesion	*Sturnus vulgaris* (Contact call) *Pheugopedius euophrys* (Contact call) *Pan troglodytes* (Pant-hoot)
2	3	3	cohesion	*Psittacus erithacus* (Contact call) *Pheugopedius euophrys* (Contact call) *Pan troglodytes* (Pant-hoot)
2	3	3	cohesion	*Pheugopedius euophrys* (Contact call) *Cercopithecus campbelli* (Boom) *Gorilla beringei* (Grunt (close call class))
2	2	11	cohesion	*Pheugopedius euophrys* (Contact call) *Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Theropithecus gelada* (Grunt) *Macaca fuscata* (Grunt) *Allochrocebus lhoesti* (Grunt) *Cercopithecus mona* (Grunt) *Erythrocebus patas* (Grunt) *Macaca mulatta* (Grunt) *Miopithecus talapoin* (Grunt)
2	2	10	cohesion	*Pheugopedius euophrys* (Contact call) *Colobus guereza* (Grunt) *Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Theropithecus gelada* (Grunt) *Semnopithecus spp. / Trachypithecus spp.* (Grunt) *Allochrocebus lhoesti* (Grunt) *Piliocolobus spp.* (Grunt) *Miopithecus talapoin* (Grunt) *Chlorocebus pygerythrus* (Grunt)
2	2	9	cohesion	*Lonchura striata domestica* (Tet call) *Colobus guereza* (Grunt) *Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Semnopithecus spp. / Trachypithecus spp.* (Grunt) *Allochrocebus lhoesti* (Grunt) *Piliocolobus spp.* (Grunt) *Miopithecus talapoin* (Grunt) *Chlorocebus pygerythrus* (Grunt)

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Classes	Families	Species	Category	Members
2	2	9	cohesion	*Pheugopedius euophrys* (Contact call) *Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Theropithecus gelada* (Grunt) *Allochrocebus lhoesti* (Grunt) *Erythrocebus patas* (Grunt) *Miopithecus talapoin* (Grunt) *Chlorocebus pygerythrus* (Grunt)
2	2	8	cohesion	*Lonchura striata domestica* (Tet call) *Cercopithecus neglectus* (Grunt) *Mandrillus leucophaeus* (Grunt) *Allochrocebus lhoesti* (Grunt) *Mandrillus sphinx* (Grunt) *Erythrocebus patas* (Grunt) *Chlorocebus pygerythrus* (Grunt) *Papio cynocephalus* (Grunt)
2	2	8	cohesion	*Lonchura striata domestica* (Tet call) *Colobus guereza* (Grunt) *Cercopithecus neglectus* (Grunt) *Semnopithecus spp. / Trachypithecus spp.* (Grunt) *Allochrocebus lhoesti* (Grunt) *Piliocolobus spp.* (Grunt) *Chlorocebus pygerythrus* (Grunt) *Papio cynocephalus* (Grunt)
2	2	8	cohesion	*Lonchura striata domestica* (Tet call) *Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Mandrillus leucophaeus* (Grunt) *Allochrocebus lhoesti* (Grunt) *Mandrillus sphinx* (Grunt) *Erythrocebus patas* (Grunt) *Chlorocebus pygerythrus* (Grunt)
2	2	8	cohesion	*Parus minor* (Contact call) *Cercopithecus neglectus* (Boom) *Mandrillus leucophaeus* (Male loud call) *Allochrocebus lhoesti* (Boom) *Mandrillus sphinx* (Male loud call) *Cercopithecus mona* (Boom) *Papio anubis* (Male loud call) *Erythrocebus patas* (Male loud call)
2	2	7	cohesion	*Lonchura striata domestica* (Tet call) *Taeniopygia guttata* (Tet call) *Colobus guereza* (Grunt) *Semnopithecus spp. / Trachypithecus spp.* (Grunt) *Piliocolobus spp.* (Grunt) *Chlorocebus pygerythrus* (Grunt) *Papio cynocephalus* (Grunt)

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Classes	Families	Species	Category	Members
2	2	7	cohesion	*Lonchura striata domestica* (Tet call) *Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Allochrocebus lhoesti* (Grunt) *Erythrocebus patas* (Grunt) *Miopithecus talapoin* (Grunt) *Chlorocebus pygerythrus* (Grunt)
2	2	7	cohesion	*Lonchura striata domestica* (Tet call) *Taeniopygia guttata* (Tet call) *Colobus guereza* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Semnopithecus spp. / Trachypithecus spp.* (Grunt) *Piliocolobus spp.* (Grunt) *Chlorocebus pygerythrus* (Grunt)
2	2	6	cohesion	*Lithobates catesbeianus* (Advertisement call) *Cercopithecus neglectus* (Boom) *Semnopithecus spp. / Trachypithecus spp.* (Loud call) *Allochrocebus lhoesti* (Boom) *Erythrocebus patas* (Male loud call) *Piliocolobus spp.* (Roar)
2	2	6	cohesion	*Lithobates catesbeianus* (Advertisement call) *Colobus guereza* (Roar) *Cercopithecus neglectus* (Boom) *Semnopithecus spp. / Trachypithecus spp.* (Loud call) *Allochrocebus lhoesti* (Boom) *Piliocolobus spp.* (Roar)
2	2	6	cohesion	*Lithobates catesbeianus* (Advertisement call) *Cercopithecus neglectus* (Boom) *Semnopithecus spp. / Trachypithecus spp.* (Loud call) *Allochrocebus lhoesti* (Boom) *Cercopithecus mona* (Boom) *Erythrocebus patas* (Male loud call)
2	2	6	cohesion	*Lithobates catesbeianus* (Advertisement call) *Colobus guereza* (Roar) *Cercopithecus neglectus* (Boom) *Semnopithecus spp. / Trachypithecus spp.* (Loud call) *Allochrocebus lhoesti* (Boom) *Cercopithecus mona* (Boom)
2	2	5	danger	*Plocepasser mahali* (Alarm call) *Macaca fuscata* (Bark) *Semnopithecus spp. / Trachypithecus spp.* (Alarm call) *Cercopithecus mona* (Alarm bark) *Macaca mulatta* (Bark)
2	2	5	danger	*Pomatostomus ruficeps* (Alarm call) *Papio ursinus* (Bark) *Macaca fuscata* (Bark) *Macaca mulatta* (Bark) *Papio cynocephalus* (Bark)
2	2	4	cohesion	*Lonchura striata domestica* (Tet call) *Taeniopygia guttata* (Tet call) *Papio ursinus* (Male loud call) *Cercopithecus nictitans* (Contact call (soft contact call))

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Classes	Families	Species	Category	Members
2	2	4	cohesion	*Parus minor* (Contact call) *Cercopithecus neglectus* (Boom) *Macaca fuscata* (Coo) *Macaca mulatta* (Coo)
2	2	4	distress	*Lonchura striata domestica* (Distress call) *Pan paniscus* (Yelp) *Gorilla beringei* (Scream) *Gorilla gorilla* (Scream (loud scream))
2	2	4	distress	*Lonchura striata domestica* (Distress call) *Taeniopygia guttata* (Distress call) *Gorilla beringei* (Scream) *Gorilla gorilla* (Scream (loud scream))
2	2	3	cohesion	*Poecile carolinensis* (Contact call) *Parus minor* (Contact call) *Cercopithecus neglectus* (Boom)
2	2	3	social	*Dendrobatidae; various genera* (Advertisement call) *Theropithecus gelada* (Copulation call) *Papio cynocephalus* (Copulation call)
2	2	3	cohesion	*Lonchura striata domestica* (Distance call) *Taeniopygia guttata* (Distance call) *Symphalangus syndactylus* (Boom)
2	2	3	cohesion	*Lonchura striata domestica* (Distance call) *Taeniopygia guttata* (Distance call) *Microcebus spp.* (Audible contact call)
2	2	3	cohesion	*Lonchura striata domestica* (Tet call) *Cercopithecus nictitans* (Contact call (soft contact call)) *Miopithecus talapoin* (Grunt)
2	2	3	danger	*Lonchura striata domestica* (Stack call) *Taeniopygia guttata* (Stack call) *Chlorocebus pygerythrus* (Chutter)
2	2	3	cohesion	*Pheugopedius euophrys* (Contact call) *Cercopithecus nictitans* (Contact call (soft contact call)) *Miopithecus talapoin* (Grunt)
2	2	3	cohesion	*Pheugopedius euophrys* (Contact call) *Papio ursinus* (Male loud call) *Cercopithecus nictitans* (Contact call (soft contact call))
2	2	3	cohesion	*Pheugopedius euophrys* (Contact call) *Cercopithecus campbelli* (Boom) *Cercopithecus ascanius* (Boom)
2	2	3	danger	*Psittacus erithacus* (Alarm call) *Cercopithecus mona* (Alarm bark) *Erythrocebus patas* (Loud bark)
2	2	2	cohesion	*Lithobates catesbeianus* (Territorial challenge call) *Poecile carolinensis* (Contact call)
2	2	2	social	*Lithobates catesbeianus* (Encounter aggression call) *Suricata suricatta* (Aggressive growl)
2	2	2	social	*Pheugopedius euophrys* (Coordinated duet song) *Crocuta crocuta* (Giggle / laugh)
2	2	2	social	*Dendrobatidae; various genera* (Advertisement call) *Cercopithecus ascanius* (Grunt)

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Classes	Families	Species	Category	Members
2	2	2	social	*Dendrobatidae; various genera* (Advertisement call) *Rousettus aegyptiacus* (Courtship / mating call)
2	2	2	social	*Dendrobatidae; various genera* (Advertisement call) *Gorilla gorilla* (Copulation grunt (soft copulation grunt))
2	2	2	social	*Dendrobatidae; various genera* (Advertisement call) *Gorilla beringei* (Hoot series / loud hoots (broadcast))
2	2	2	social	*Dendrobatidae; various genera* (Advertisement call) *Pongo pygmaeus* (Long call)
2	2	2	social	*Dendrobatidae; various genera* (Advertisement call) *Symphalangus syndactylus* (Boom)
2	2	2	social	*Dendrobatidae; various genera* (Aggressive call) *Pan troglodytes* (Waa-bark)
2	2	2	social	*Engystomops pustulosus* (Whine–chuck advertisement call) *Pongo pygmaeus* (Long call)
2	2	2	social	*Engystomops pustulosus* (Whine–chuck advertisement call) *Alouatta spp.* (Grunt)
2	2	2	social	*Engystomops pustulosus* (Whine–chuck advertisement call) *Callithrix jacchus* (Peep)
2	2	2	social	*Engystomops pustulosus* (Aggressive call) *Symphalangus syndactylus* (Great call (female-led duet))
2	2	2	cohesion	*Engystomops pustulosus* (Release call) *Plocepasser mahali* (Alarm call)
2	2	2	cohesion	*Pheugopedius euophrys* (Coordinated duet song) *Canis lupus* (Howl)
2	2	2	social	*Pheugopedius euophrys* (Coordinated duet song) *Saccopteryx bilineata* (Territorial song)
2	2	2	cohesion	*Plocepasser mahali* (Duet-like cooperative call) *Hylobates spp., Nomascus spp., Hoolock spp.* (Great call / song bout)
2	2	2	cohesion	*Sturnus vulgaris* (Contact call) *Indri indri* (Contact call)
2	2	2	danger	*Psittacus erithacus* (Alarm call) *Cercopithecus pogonias* (Hack (sharp warning/alarm call))
2	2	2	cohesion	*Hyla cinerea* (Aggressive call) *Cercopithecus ascanius* (Boom)
2	2	2	social	*Hyla cinerea* (Advertisement call) *Theropithecus gelada* (Copulation call)

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Classes	Families	Species	Category	Members
1	6	26	distress	*Colobus guereza* (Infant distress call) *Eulemur spp.* (Infant distress call) *Papio ursinus* (Infant distress call) *Callithrix jacchus* (Infant distress call) *Cercopithecus pogonias* (Infant distress call) *Cercopithecus neglectus* (Infant distress call) *Cercopithecus diana* (Infant distress call) *Mandrillus leucophaeus* (Infant distress call) *Theropithecus gelada* (Infant distress call) *Indri indri* (Infant distress call) *Macaca fuscata* (Infant distress call) *Semnopithecus spp. / Trachypithecus spp.* (Infant distress call) *Allochrocebus lhoesti* (Infant distress call) *Mandrillus sphinx* (Infant distress call) *Cercopithecus mona* (Infant distress call) *Microcebus spp.* (Infant distress call) *Erythrocebus patas* (Infant distress call) *Cercopithecus nictitans* (Infant distress call) *Piliocolobus spp.* (Infant distress call) *Cercopithecus ascanius* (Infant distress call) *Macaca mulatta* (Infant distress call) *Lemur catta* (Infant distress call) *Miopithecus talapoin* (Infant distress call) *Sapajus apella* (Infant distress call) *Chlorocebus pygerythrus* (Infant distress call) *Papio cynocephalus* (Infant distress call)
1	5	15	distress	*Colobus guereza* (Scream) *Pongo pygmaeus* (Bared-teeth scream) *Eulemur spp.* (Scream) *Callithrix jacchus* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Theropithecus gelada* (Scream) *Alouatta spp.* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Mandrillus sphinx* (Scream) *Cercopithecus mona* (Scream) *Cercopithecus nictitans* (Scream) *Cercopithecus ascanius* (Scream) *Miopithecus talapoin* (Scream)
1	5	14	distress	*Colobus guereza* (Scream) *Eulemur spp.* (Scream) *Papio ursinus* (Scream) *Callithrix jacchus* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Saccopteryx bilineata* (Aggressive call) *Alouatta spp.* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Cercopithecus mona* (Scream) *Cercopithecus nictitans* (Scream) *Piliocolobus spp.* (Scream) *Cercopithecus ascanius* (Scream)

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Classes	Families	Species	Category	Members
1	4	17	distress	*Colobus guereza* (Scream) *Eulemur spp.* (Scream) *Papio ursinus* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Cercopithecus diana* (Scream) *Saccopteryx bilineata* (Aggressive call) *Alouatta spp.* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Cercopithecus mona* (Scream) *Erythrocebus patas* (Scream) *Cercopithecus nictitans* (Scream) *Piliocolobus spp.* (Scream) *Cercopithecus ascanius* (Scream) *Lemur catta* (Scream) *Chlorocebus pygerythrus* (Scream)
1	4	17	distress	*Colobus guereza* (Scream) *Eulemur spp.* (Scream) *Papio ursinus* (Scream) *Callithrix jacchus* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Mandrillus leucophaeus* (Scream) *Theropithecus gelada* (Scream) *Alouatta spp.* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Mandrillus sphinx* (Scream) *Cercopithecus mona* (Scream) *Cercopithecus nictitans* (Scream) *Piliocolobus spp.* (Scream) *Cercopithecus ascanius* (Scream) *Miopithecus talapoin* (Scream)
1	4	16	distress	*Colobus guereza* (Scream) *Pongo pygmaeus* (Bared-teeth scream) *Eulemur spp.* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Cercopithecus diana* (Scream) *Theropithecus gelada* (Scream) *Alouatta spp.* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Mandrillus sphinx* (Scream) *Cercopithecus mona* (Scream) *Cercopithecus nictitans* (Scream) *Cercopithecus ascanius* (Scream) *Lemur catta* (Scream) *Miopithecus talapoin* (Scream)

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Classes	Families	Species	Category	Members
1	4	13	distress	*Pan paniscus* (Scream) *Callithrix jacchus* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Mandrillus leucophaeus* (Scream) *Theropithecus gelada* (Scream) *Alouatta spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Mandrillus sphinx* (Scream) *Cercopithecus mona* (Scream) *Cercopithecus nictitans* (Scream) *Cercopithecus ascanius* (Scream) *Miopithecus talapoin* (Scream)
1	4	7	distress	*Callithrix jacchus* (Infant distress call) *Cercopithecus diana* (Infant distress call) *Indri indri* (Infant distress call) *Papio anubis* (Infant distress call) *Cercopithecus nictitans* (Infant distress call) *Lemur catta* (Infant distress call) *Papio cynocephalus* (Infant distress call)
1	4	6	danger	*Eulemur spp.* (Alarm call) *Papio ursinus* (Bark) *Indri indri* (Alarm call) *Papio anubis* (Bark) *Lemur catta* (Alarm call (general)) *Sapajus apella* (Alarm call)
1	4	5	social	*Cercopithecus pogonias* (Scream) *Rousettus aegyptiacus* (Social squabble call) *Saccopteryx bilineata* (Aggressive call) *Alouatta spp.* (Scream) *Cercopithecus ascanius* (Scream)
1	3	22	distress	*Colobus guereza* (Scream) *Eulemur spp.* (Scream) *Papio ursinus* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Cercopithecus diana* (Scream) *Mandrillus leucophaeus* (Scream) *Theropithecus gelada* (Scream) *Macaca fuscata* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Mandrillus sphinx* (Scream) *Cercopithecus mona* (Scream) *Erythrocebus patas* (Scream) *Cercopithecus nictitans* (Scream) *Piliocolobus spp.* (Scream) *Cercopithecus ascanius* (Scream) *Macaca mulatta* (Scream) *Lemur catta* (Scream) *Sapajus apella* (Scream) *Chlorocebus pygerythrus* (Scream) *Papio cynocephalus* (Scream)

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Classes	Families	Species	Category	Members
1	3	22	distress	*Colobus guereza* (Scream) *Eulemur spp.* (Scream) *Papio ursinus* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Cercopithecus diana* (Scream) *Mandrillus leucophaeus* (Scream) *Theropithecus gelada* (Scream) *Macaca fuscata* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Mandrillus sphinx* (Scream) *Cercopithecus mona* (Scream) *Erythrocebus patas* (Scream) *Cercopithecus nictitans* (Scream) *Piliocolobus spp.* (Scream) *Cercopithecus ascanius* (Scream) *Macaca mulatta* (Scream) *Lemur catta* (Scream) *Miopithecus talapoin* (Scream) *Sapajus apella* (Scream) *Chlorocebus pygerythrus* (Scream)
1	3	22	distress	*Colobus guereza* (Scream) *Eulemur spp.* (Scream) *Papio ursinus* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Cercopithecus diana* (Scream) *Mandrillus leucophaeus* (Scream) *Theropithecus gelada* (Scream) *Alouatta spp.* (Scream) *Macaca fuscata* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Mandrillus sphinx* (Scream) *Cercopithecus mona* (Scream) *Erythrocebus patas* (Scream) *Cercopithecus nictitans* (Scream) *Piliocolobus spp.* (Scream) *Cercopithecus ascanius* (Scream) *Macaca mulatta* (Scream) *Lemur catta* (Scream) *Miopithecus talapoin* (Scream) *Chlorocebus pygerythrus* (Scream)

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Classes	Families	Species	Category	Members
1	3	17	distress	*Colobus guereza* (Scream) *Papio ursinus* (Scream) *Cercopithecus pogonias* (Scream) *Cercopithecus neglectus* (Scream) *Mandrillus leucophaeus* (Scream) *Theropithecus gelada* (Scream) *Alouatta spp.* (Scream) *Semnopithecus spp. / Trachypithecus spp.* (Scream) *Allochrocebus lhoesti* (Scream) *Mandrillus sphinx* (Scream) *Cercopithecus mona* (Scream) *Erythrocebus patas* (Scream) *Piliocolobus spp.* (Scream) *Cercopithecus ascanius* (Scream) *Miopithecus talapoin* (Scream) *Chlorocebus pygerythrus* (Scream) *Gorilla gorilla* (Scream (loud scream))
1	3	13	distress	*Colobus guereza* (Infant distress call) *Eulemur spp.* (Infant distress call) *Papio ursinus* (Infant distress call) *Cercopithecus pogonias* (Infant distress call) *Cercopithecus neglectus* (Infant distress call) *Mandrillus leucophaeus* (Infant distress call) *Semnopithecus spp. / Trachypithecus spp.* (Infant distress call) *Allochrocebus lhoesti* (Infant distress call) *Mandrillus sphinx* (Infant distress call) *Suricata suricatta* (Infant begging call) *Erythrocebus patas* (Infant distress call) *Piliocolobus spp.* (Infant distress call) *Cercopithecus ascanius* (Infant distress call)
1	3	6	social	*Mandrillus leucophaeus* (Threat growl) *Macaca fuscata* (Threat growl) *Mandrillus sphinx* (Threat growl) *Suricata suricatta* (Aggressive growl) *Macaca mulatta* (Threat growl) *Sapajus apella* (Threat growl)
1	3	6	social	*Mandrillus leucophaeus* (Threat growl) *Alouatta spp.* (Wrah / Growl) *Macaca fuscata* (Threat growl) *Mandrillus sphinx* (Threat growl) *Macaca mulatta* (Threat growl) *Crocuta crocuta* (Growl)
1	3	6	cohesion	*Eulemur spp.* (Grunt) *Mandrillus leucophaeus* (Grunt) *Gorilla beringei* (Grunt (close call class)) *Mandrillus sphinx* (Grunt) *Erythrocebus patas* (Grunt) *Papio cynocephalus* (Grunt)
1	3	6	danger	*Eulemur spp.* (Alarm call) *Papio ursinus* (Bark) *Theropithecus gelada* (Alarm bark) *Indri indri* (Alarm call) *Papio anubis* (Bark) *Lemur catta* (Alarm call (general))

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Classes	Families	Species	Category	Members
1	3	5	danger	*Papio ursinus* (Bark) *Cercopithecus pogonias* (Hack (sharp warning/alarm call)) *Indri indri* (Alarm call) *Papio anubis* (Bark) *Lemur catta* (Alarm call (general))
1	3	4	cohesion	*Pan troglodytes* (Pant-hoot) *Cercopithecus neglectus* (Boom) *Allochrocebus lhoesti* (Boom) *Crocuta crocuta* (Whoop)
1	3	4	cohesion	*Colobus guereza* (Grunt) *Eulemur spp.* (Grunt) *Gorilla beringei* (Grunt (close call class)) *Papio cynocephalus* (Grunt)
1	3	4	cohesion	*Colobus guereza* (Grunt) *Semnopithecus spp. / Trachypithecus spp.* (Grunt) *Symphalangus syndactylus* (Male answer phrase) *Sapajus apella* (Contact call (chirp / peep))
1	3	4	cohesion	*Pan troglodytes* (Pant-hoot) *Cercopithecus neglectus* (Boom) *Hylobates spp., Nomascus spp., Hoolock spp.* (Male solo song) *Allochrocebus lhoesti* (Boom)
1	3	4	cohesion	*Pan paniscus* (Whistle) *Cercopithecus neglectus* (Boom) *Allochrocebus lhoesti* (Boom) *Microcebus spp.* (Audible contact call)
1	3	4	danger	*Eulemur spp.* (Alarm call) *Indri indri* (Alarm call) *Microcebus spp.* (Alarm call) *Lemur catta* (Alarm call (general))
1	3	4	danger	*Eulemur spp.* (Alarm call) *Cercopithecus campbelli* (Krak) *Lemur catta* (Alarm call (general)) *Sapajus apella* (Alarm call)
1	3	3	cohesion	*Poecile carolinensis* (Contact call) *Pomatostomus ruficeps* (Flight call) *Plocepasser mahali* (Contact call)
1	3	3	danger	*Cercopithecus campbelli* (Krak) *Pan troglodytes* (Alarm huu) *Suricata suricatta* (Terrestrial predator alarm call)
1	3	3	social	*Suricata suricatta* (Aggressive growl) *Sapajus apella* (Threat growl) *Papio cynocephalus* (Threat grunt / growl)
1	3	3	distress	*Rousettus aegyptiacus* (Mother–infant call) *Saccopteryx bilineata* (Mother–infant isolation call) *Suricata suricatta* (Infant begging call)
1	3	3	distress	*Saccopteryx bilineata* (Mother–infant isolation call) *Microcebus spp.* (Infant distress call) *Crocuta crocuta* (Cub distress call)
1	3	3	social	*Hyla cinerea* (Aggressive call) *Dendrobatidae; various genera* (Aggressive call) *Engystomops pustulosus* (Aggressive call)

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Classes	Families	Species	Category	Members
1	3	3	cohesion	*Lithobates catesbeianus* (Release call) *Hyla cinerea* (Release call) *Engystomops pustulosus* (Release call)
1	3	3	cohesion	*Pan troglodytes* (Pant-hoot) *Canis lupus* (Howl) *Indri indri* (Song (group song / chorus))
1	3	3	cohesion	*Pan troglodytes* (Pant-hoot) *Hylobates spp., Nomascus spp., Hoolock spp.* (Male solo song) *Canis lupus* (Howl)
1	3	3	cohesion	*Colobus guereza* (Grunt) *Pan paniscus* (Peep) *Sapajus apella* (Contact call (chirp / peep))
1	3	3	other	*Pan troglodytes* (Grunt) *Lemur catta* (Contact call (grunt / short contact call)) *Sapajus apella* (Contact call (chirp / peep))
1	3	3	cohesion	*Hylobates spp., Nomascus spp., Hoolock spp.* (Soft grunt / contact call) *Alouatta spp.* (Grunt) *Gorilla gorilla* (Single grunt)
1	3	3	other	*Lemur catta* (Contact call (grunt / short contact call)) *Symphalangus syndactylus* (Male answer phrase) *Sapajus apella* (Contact call (chirp / peep))
1	3	3	cohesion	*Eulemur spp.* (Grunt) *Hylobates spp., Nomascus spp., Hoolock spp.* (Soft grunt / contact call) *Alouatta spp.* (Grunt)
1	2	12	cohesion	*Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Mandrillus leucophaeus* (Grunt) *Gorilla beringei* (Grunt (close call class)) *Macaca fuscata* (Grunt) *Allochrocebus lhoesti* (Grunt) *Mandrillus sphinx* (Grunt) *Cercopithecus mona* (Grunt) *Erythrocebus patas* (Grunt) *Cercopithecus ascanius* (Grunt) *Macaca mulatta* (Grunt) *Papio cynocephalus* (Grunt)
1	2	12	cohesion	*Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Mandrillus leucophaeus* (Grunt) *Gorilla beringei* (Grunt (close call class)) *Macaca fuscata* (Grunt) *Allochrocebus lhoesti* (Grunt) *Mandrillus sphinx* (Grunt) *Cercopithecus mona* (Grunt) *Erythrocebus patas* (Grunt) *Cercopithecus ascanius* (Grunt) *Macaca mulatta* (Grunt)

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Classes	Families	Species	Category	Members
1	2	11	cohesion	*Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Gorilla beringei* (Grunt (close call class)) *Macaca fuscata* (Grunt) *Allochrocebus lhoesti* (Grunt) *Cercopithecus mona* (Grunt) *Erythrocebus patas* (Grunt) *Cercopithecus ascanius* (Grunt) *Macaca mulatta* (Grunt) *Miopithecus talapoin* (Grunt)
1	2	10	cohesion	*Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Mandrillus leucophaeus* (Grunt) *Gorilla beringei* (Grunt (close call class)) *Allochrocebus lhoesti* (Grunt) *Mandrillus sphinx* (Grunt) *Erythrocebus patas* (Grunt) *Cercopithecus ascanius* (Grunt) *Chlorocebus pygerythrus* (Grunt) *Papio cynocephalus* (Grunt)
1	2	10	cohesion	*Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Mandrillus leucophaeus* (Grunt) *Gorilla beringei* (Grunt (close call class)) *Allochrocebus lhoesti* (Grunt) *Mandrillus sphinx* (Grunt) *Erythrocebus patas* (Grunt) *Cercopithecus ascanius* (Grunt) *Chlorocebus pygerythrus* (Grunt)
1	2	9	cohesion	*Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Gorilla beringei* (Grunt (close call class)) *Allochrocebus lhoesti* (Grunt) *Erythrocebus patas* (Grunt) *Cercopithecus ascanius* (Grunt) *Miopithecus talapoin* (Grunt) *Chlorocebus pygerythrus* (Grunt)
1	2	8	cohesion	*Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Mandrillus leucophaeus* (Grunt) *Allochrocebus lhoesti* (Grunt) *Mandrillus sphinx* (Grunt) *Erythrocebus patas* (Grunt) *Cercopithecus ascanius* (Grunt) *Crocuta crocuta* (Groan)

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Classes	Families	Species	Category	Members
1	2	8	social	*Pan paniscus* (Whistle) *Papio ursinus* (Male loud call) *Cercopithecus neglectus* (Boom) *Mandrillus leucophaeus* (Male loud call) *Allochrocebus lhoesti* (Boom) *Mandrillus sphinx* (Male loud call) *Papio anubis* (Male loud call) *Erythrocebus patas* (Male loud call)
1	2	7	social	*Papio ursinus* (Male loud call) *Mandrillus leucophaeus* (Male loud call) *Mandrillus sphinx* (Male loud call) *Suricata suricatta* (Close-range contact call) *Cercopithecus mona* (Boom) *Papio anubis* (Male loud call) *Erythrocebus patas* (Male loud call)
1	2	7	cohesion	*Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Allochrocebus lhoesti* (Grunt) *Erythrocebus patas* (Grunt) *Cercopithecus ascanius* (Grunt) *Crocuta crocuta* (Groan) *Miopithecus talapoin* (Grunt)
1	2	7	danger	*Eulemur spp.* (Alarm call) *Mandrillus leucophaeus* (Alarm bark) *Theropithecus gelada* (Alarm bark) *Semnopithecus spp. / Trachypithecus spp.* (Alarm call) *Mandrillus sphinx* (Alarm bark) *Cercopithecus mona* (Alarm bark) *Erythrocebus patas* (Loud bark)
1	2	6	cohesion	*Eulemur spp.* (Grunt) *Mandrillus leucophaeus* (Grunt) *Mandrillus sphinx* (Grunt) *Papio anubis* (Grunt) *Erythrocebus patas* (Grunt) *Papio cynocephalus* (Grunt)
1	2	6	danger	*Eulemur spp.* (Alarm call) *Papio ursinus* (Bark) *Mandrillus leucophaeus* (Alarm bark) *Theropithecus gelada* (Alarm bark) *Mandrillus sphinx* (Alarm bark) *Papio anubis* (Bark)
1	2	6	danger	*Eulemur spp.* (Alarm call) *Papio ursinus* (Bark) *Mandrillus leucophaeus* (Alarm bark) *Theropithecus gelada* (Alarm bark) *Mandrillus sphinx* (Alarm bark) *Cercopithecus mona* (Alarm bark)
1	2	5	cohesion	*Cercopithecus neglectus* (Boom) *Semnopithecus spp. / Trachypithecus spp.* (Loud call) *Allochrocebus lhoesti* (Boom) *Cercopithecus mona* (Boom) *Crocuta crocuta* (Whoop)

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Classes	Families	Species	Category	Members
1	2	5	social	*Macaca fuscata* (Copulation scream) *Papio anubis* (Copulation call) *Macaca mulatta* (Copulation scream) *Sapajus apella* (Copulation call) *Papio cynocephalus* (Copulation call)
1	2	4	cohesion	*Papio ursinus* (Grunt) *Mandrillus leucophaeus* (Grunt) *Mandrillus sphinx* (Grunt) *Crocuta crocuta* (Groan)
1	2	4	danger	*Eulemur spp.* (Alarm call) *Cercopithecus campbelli* (Krak) *Mandrillus sphinx* (Alarm bark) *Cercopithecus mona* (Alarm bark)
1	2	4	cohesion	*Colobus guereza* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Semnopithecus spp. / Trachypithecus spp.* (Grunt) *Symphalangus syndactylus* (Male answer phrase)
1	2	4	cohesion	*Pan troglodytes* (Pant-hoot) *Cercopithecus neglectus* (Boom) *Allochrocebus lhoesti* (Boom) *Cercopithecus ascanius* (Boom)
1	2	4	danger	*Eulemur spp.* (Alarm call) *Cercopithecus campbelli* (Krak) *Cercopithecus mona* (Alarm bark) *Lemur catta* (Alarm call (general))
1	2	4	danger	*Papio ursinus* (Bark) *Indri indri* (Alarm call) *Macaca fuscata* (Bark) *Macaca mulatta* (Bark)
1	2	4	danger	*Papio ursinus* (Bark) *Indri indri* (Alarm call) *Papio anubis* (Bark) *Chlorocebus pygerythrus* (Eagle alarm call)
1	2	4	cohesion	*Papio anubis* (Grunt) *Lemur catta* (Contact call (grunt / short contact call)) *Chlorocebus pygerythrus* (Grunt) *Papio cynocephalus* (Grunt)
1	2	3	cohesion	*Pan paniscus* (Laughter) *Crocuta crocuta* (Giggle / laugh) *Gorilla gorilla* (Play chuckle (rare in wild recordings))
1	2	3	cohesion	*Loxodonta africana* (Rumble (contact / coordination; incl. infrasonic rumbles)) *Mandrillus leucophaeus* (Grunt) *Mandrillus sphinx* (Grunt)
1	2	3	distress	*Loxodonta africana* (Trumpet) *Papio ursinus* (Scream) *Mandrillus sphinx* (Scream)
1	2	3	social	*Mandrillus leucophaeus* (Copulation call) *Saccopteryx bilineata* (Courtship song) *Mandrillus sphinx* (Copulation call)

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Classes	Families	Species	Category	Members
1	2	3	cohesion	*Psittacus erithacus* (Contact call) *Lonchura striata domestica* (Distance call) *Taeniopygia guttata* (Distance call)
1	2	3	cohesion	*Colobus guereza* (Grunt) *Microcebus spp.* (Audible contact call) *Papio cynocephalus* (Grunt)
1	2	3	distress	*Pan troglodytes* (Scream) *Cercopithecus diana* (Scream) *Cercopithecus nictitans* (Scream)
1	2	3	distress	*Pan paniscus* (Yelp) *Gorilla beringei* (Scream) *Hylobates spp., Nomascus spp., Hoolock spp.* (Alarm scream)
1	2	3	social	*Papio anubis* (Threat grunt / growl) *Gorilla gorilla* (Soft threat grunt) *Papio cynocephalus* (Threat grunt / growl)
1	2	3	danger	*Pongo pygmaeus* (Bark) *Chlorocebus pygerythrus* (Bark) *Papio cynocephalus* (Bark)
1	2	3	cohesion	*Allochrocebus lhoesti* (Boom) *Cercopithecus mona* (Boom) *Symphalangus syndactylus* (Boom)
1	2	3	cohesion	*Callithrix jacchus* (Peep) *Macaca fuscata* (Grunt) *Macaca mulatta* (Grunt)
1	2	3	social	*Theropithecus gelada* (Copulation call) *Sapajus apella* (Copulation call) *Papio cynocephalus* (Copulation call)
1	2	2	danger	*Poecile carolinensis* (Seet call) *Sturnus vulgaris* (Alarm call)
1	2	2	danger	*Callithrix jacchus* (Tsik (alarm call)) *Suricata suricatta* (Aerial predator alarm call)
1	2	2	danger	*Suricata suricatta* (Aerial predator alarm call) *Chlorocebus pygerythrus* (Eagle alarm call)
1	2	2	danger	*Suricata suricatta* (Terrestrial predator alarm call) *Microcebus spp.* (Alarm call)
1	2	2	cohesion	*Pan troglodytes* (Waa-bark) *Suricata suricatta* (Recruitment call)
1	2	2	social	*Canis lupus* (Whine) *Crocuta crocuta* (Whine)
1	2	2	social	*Lithobates catesbeianus* (Territorial challenge call) *Dendrobatidae; various genera* (Aggressive call)
1	2	2	cohesion	*Loxodonta africana* (Rumble (contact / coordination; incl. infrasonic rumbles)) *Hylobates spp., Nomascus spp., Hoolock spp.* (Soft grunt / contact call)
1	2	2	cohesion	*Callithrix jacchus* (Trill) *Rousettus aegyptiacus* (Affiliative contact call)
1	2	2	social	*Pongo pygmaeus* (Mating squeals) *Rousettus aegyptiacus* (Courtship / mating call)

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Classes	Families	Species	Category	Members
1	2	2	cohesion	*Rousettus aegyptiacus* (Group coordination call) *Papio anubis* (Male loud call)
1	2	2	social	*Saccopteryx bilineata* (Territorial song) *Indri indri* (Song (group song / chorus))
1	2	2	social	*Lonchura striata domestica* (Male song) *Sturnus vulgaris* (Male song (long bout))
1	2	2	social	*Lonchura striata domestica* (Male song) *Pheugopedius euophrys* (Coordinated duet song)
1	2	2	social	*Pheugopedius euophrys* (Coordinated duet song) *Taeniopygia guttata* (Directed song)
1	2	2	social	*Sturnus vulgaris* (Male song (long bout)) *Taeniopygia guttata* (Directed song)
1	2	2	social	*Mandrillus leucophaeus* (Copulation call) *Gorilla gorilla* (Copulation grunt (soft copulation grunt))
1	2	2	social	*Pan troglodytes* (Laughter (play laughter)) *Mandrillus leucophaeus* (Copulation call)
1	2	2	cohesion	*Pan troglodytes* (Pant-hoot) *Papio cynocephalus* (Wahoo)
1	2	2	danger	*Pan troglodytes* (Alarm huu) *Allochrocebus lhoesti* (Leopard alarm call)
1	2	2	cohesion	*Symphalangus syndactylus* (Great call (female-led duet)) *Gorilla gorilla* (Single grunt)
1	2	2	distress	*Pongo pygmaeus* (Soft hoot) *Alouatta spp.* (Infant distress call)
1	2	2	distress	*Pongo pygmaeus* (Soft hoot) *Microcebus spp.* (Infant distress call)
1	2	2	danger	*Theropithecus gelada* (Alarm bark) *Hylobates spp., Nomascus spp., Hoolock spp.* (Alarm scream)
1	2	2	danger	*Cercopithecus campbelli* (Krak) *Hylobates spp., Nomascus spp., Hoolock spp.* (Alarm scream)
1	2	2	distress	*Alouatta spp.* (Infant distress call) *Indri indri* (Infant distress call)
1	2	2	danger	*Sapajus apella* (Bark) *Chlorocebus pygerythrus* (Bark)
1	2	2	social	*Alouatta spp.* (Wrah / Growl) *Papio cynocephalus* (Threat grunt / growl)
1	2	2	other	*Callithrix jacchus* (Peep) *Miopithecus talapoin* (Chirp)

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Classes	Families	Species	Category	Members
1	1	12	cohesion	*Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Mandrillus leucophaeus* (Grunt) *Macaca fuscata* (Grunt) *Allochrocebus lhoesti* (Grunt) *Mandrillus sphinx* (Grunt) *Cercopithecus mona* (Grunt) *Papio anubis* (Grunt) *Erythrocebus patas* (Grunt) *Cercopithecus ascanius* (Grunt) *Macaca mulatta* (Grunt) *Papio cynocephalus* (Grunt)
1	1	11	cohesion	*Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Theropithecus gelada* (Grunt) *Macaca fuscata* (Grunt) *Allochrocebus lhoesti* (Grunt) *Cercopithecus mona* (Grunt) *Erythrocebus patas* (Grunt) *Cercopithecus ascanius* (Grunt) *Macaca mulatta* (Grunt) *Miopithecus talapoin* (Grunt)
1	1	10	cohesion	*Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Mandrillus leucophaeus* (Grunt) *Allochrocebus lhoesti* (Grunt) *Mandrillus sphinx* (Grunt) *Papio anubis* (Grunt) *Erythrocebus patas* (Grunt) *Cercopithecus ascanius* (Grunt) *Chlorocebus pygerythrus* (Grunt) *Papio cynocephalus* (Grunt)
1	1	9	cohesion	*Papio ursinus* (Male loud call) *Cercopithecus pogonias* (Boom) *Cercopithecus neglectus* (Boom) *Mandrillus leucophaeus* (Male loud call) *Allochrocebus lhoesti* (Boom) *Mandrillus sphinx* (Male loud call) *Cercopithecus mona* (Boom) *Papio anubis* (Male loud call) *Erythrocebus patas* (Male loud call)
1	1	9	cohesion	*Papio ursinus* (Male loud call) *Cercopithecus neglectus* (Boom) *Mandrillus leucophaeus* (Male loud call) *Semnopithecus spp. / Trachypithecus spp.* (Loud call) *Allochrocebus lhoesti* (Boom) *Mandrillus sphinx* (Male loud call) *Papio anubis* (Male loud call) *Erythrocebus patas* (Male loud call) *Piliocolobus spp.* (Roar)

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Classes	Families	Species	Category	Members
1	1	9	cohesion	*Papio ursinus* (Male loud call) *Cercopithecus neglectus* (Boom) *Mandrillus leucophaeus* (Male loud call) *Semnopithecus spp. / Trachypithecus spp.* (Loud call) *Allochrocebus lhoesti* (Boom) *Mandrillus sphinx* (Male loud call) *Cercopithecus mona* (Boom) *Papio anubis* (Male loud call) *Erythrocebus patas* (Male loud call)
1	1	9	cohesion	*Colobus guereza* (Roar) *Papio ursinus* (Male loud call) *Cercopithecus neglectus* (Boom) *Mandrillus leucophaeus* (Male loud call) *Semnopithecus spp. / Trachypithecus spp.* (Loud call) *Allochrocebus lhoesti* (Boom) *Mandrillus sphinx* (Male loud call) *Papio anubis* (Male loud call) *Piliocolobus spp.* (Roar)
1	1	9	cohesion	*Colobus guereza* (Roar) *Papio ursinus* (Male loud call) *Cercopithecus neglectus* (Boom) *Mandrillus leucophaeus* (Male loud call) *Semnopithecus spp. / Trachypithecus spp.* (Loud call) *Allochrocebus lhoesti* (Boom) *Mandrillus sphinx* (Male loud call) *Cercopithecus mona* (Boom) *Papio anubis* (Male loud call)
1	1	9	cohesion	*Cercopithecus pogonias* (Grunt) *Cercopithecus neglectus* (Grunt) *Cercopithecus diana* (Contact call (soft contact call)) *Theropithecus gelada* (Grunt) *Allochrocebus lhoesti* (Grunt) *Erythrocebus patas* (Grunt) *Cercopithecus ascanius* (Grunt) *Miopithecus talapoin* (Grunt) *Chlorocebus pygerythrus* (Grunt)
1	1	8	cohesion	*Papio ursinus* (Grunt) *Cercopithecus pogonias* (Grunt) *Mandrillus leucophaeus* (Grunt) *Macaca fuscata* (Grunt) *Mandrillus sphinx* (Grunt) *Cercopithecus mona* (Grunt) *Macaca mulatta* (Grunt) *Papio cynocephalus* (Grunt)
1	1	5	social	*Papio ursinus* (Copulation call) *Mandrillus leucophaeus* (Copulation call) *Theropithecus gelada* (Copulation call) *Mandrillus sphinx* (Copulation call) *Papio anubis* (Copulation call)
1	1	5	danger	*Macaca fuscata* (Bark) *Papio anubis* (Bark) *Macaca mulatta* (Bark) *Chlorocebus pygerythrus* (Bark) *Papio cynocephalus* (Bark)

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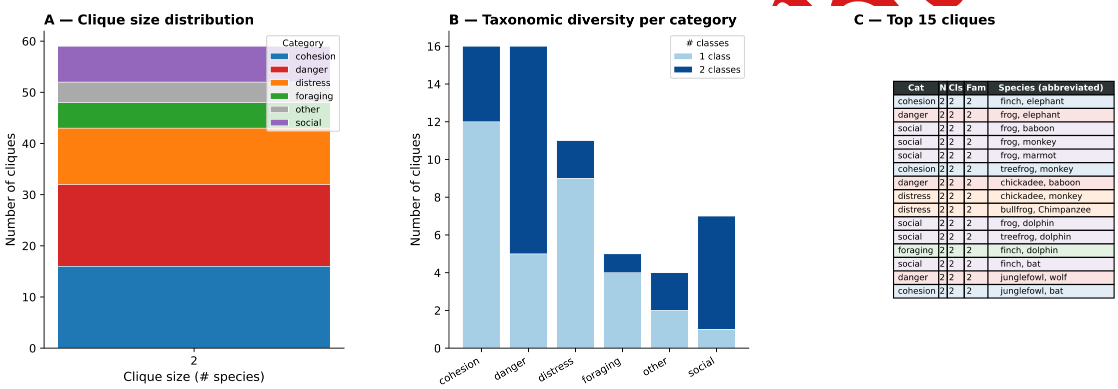
Classes	Families	Species	Category	Members
1	1	4	danger	*Cercopithecus neglectus* (Eagle alarm call) *Cercopithecus diana* (Eagle alarm call) *Allochrocebus lhoesti* (Eagle alarm call) *Erythrocebus patas* (Eagle alarm call)
1	1	3	distress	*Mandrillus leucophaeus* (Infant distress call) *Mandrillus sphinx* (Infant distress call) *Papio anubis* (Infant distress call)
1	1	3	cohesion	*Pan paniscus* (Laughter) *Pan troglodytes* (Laughter (play laughter)) *Gorilla gorilla* (Play chuckle (rare in wild recordings))
1	1	3	danger	*Cercopithecus campbelli* (Krak) *Cercopithecus nictitans* (Pyow) *Cercopithecus ascanius* (Leopard alarm call)
1	1	3	cohesion	*Papio ursinus* (Male loud call) *Cercopithecus mona* (Boom) *Cercopithecus nictitans* (Contact call (soft contact call))
1	1	3	danger	*Cercopithecus neglectus* (Leopard alarm call) *Allochrocebus lhoesti* (Leopard alarm call) *Cercopithecus ascanius* (Leopard alarm call)
1	1	3	danger	*Cercopithecus neglectus* (Snake alarm call) *Allochrocebus lhoesti* (Snake alarm call) *Cercopithecus ascanius* (Snake alarm call)
1	1	2	cohesion	*Piliocolobus spp.* (Roar) *Miopithecus talapoin* (Contact twitter)
1	1	2	danger	*Colobus guereza* (Snort) *Piliocolobus spp.* (Snort)
1	1	2	cohesion	*Pan paniscus* (Pant grunt) *Pan troglodytes* (Pant-grunt)
1	1	2	social	*Pongo pygmaeus* (Mating squeals) *Gorilla gorilla* (Copulation grunt (soft copulation grunt))
1	1	2	cohesion	*Gorilla beringei* (Hoot series / loud hoots (broadcast)) *Gorilla gorilla* (Hoot series (long-distance call))
1	1	2	cohesion	*Hylobates spp., Nomascus spp., Hoolock spp.* (Soft grunt / contact call) *Symphalangus syndactylus* (Soft hoot)
1	1	2	cohesion	*Macaca fuscata* (Girney) *Macaca mulatta* (Girney)
1	1	2	danger	*Cercopithecus campbelli* (Hok) *Cercopithecus nictitans* (Hack)
1	1	2	distress	*Papio anubis* (Scream) *Papio cynocephalus* (Scream)
1	1	2	cohesion	*Papio anubis* (Wahoo) *Papio cynocephalus* (Wahoo)



Cat	N	Cls	Fam	Species (abbreviated)
cohesion	2	2	2	finch, elephant
danger	2	2	2	frog, elephant
social	2	2	2	frog, baboon
social	2	2	2	frog, monkey
social	2	2	2	frog, marmot
cohesion	2	2	2	treefrog, monkey
danger	2	2	2	chickadee, baboon
distress	2	2	2	chickadee, monkey
distress	2	2	2	bullfrog, Chimpanzee
social	2	2	2	frog, dolphin
social	2	2	2	treefrog, dolphin
foraging	2	2	2	finch, dolphin
social	2	2	2	finch, bat
danger	2	2	2	junglefowl, wolf
cohesion	2	2	2	junglefowl, bat

Figure 11. **Distribution of cross-species call groups.** *Left*: number of groups by size, coloured by dominant semantic category. *Centre*: number of groups with 1, 2, or 3 taxonomic classes, grouped by category. *Right*: top-15 groups by taxonomic diversity.

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Figure 12, Figure 13, Figure 14, Figure 15.

Figure 12. Acoustic embedding space (UMAP). Each point is a call in AnimalLex, positioned by acoustic description embedding and colored by taxonomic class. Clusters correspond to spectro-temporal call types (tonal long calls, broadband alarm bursts, etc.) that recur across classes, illustrating the convergent acoustic solutions captured by the database.

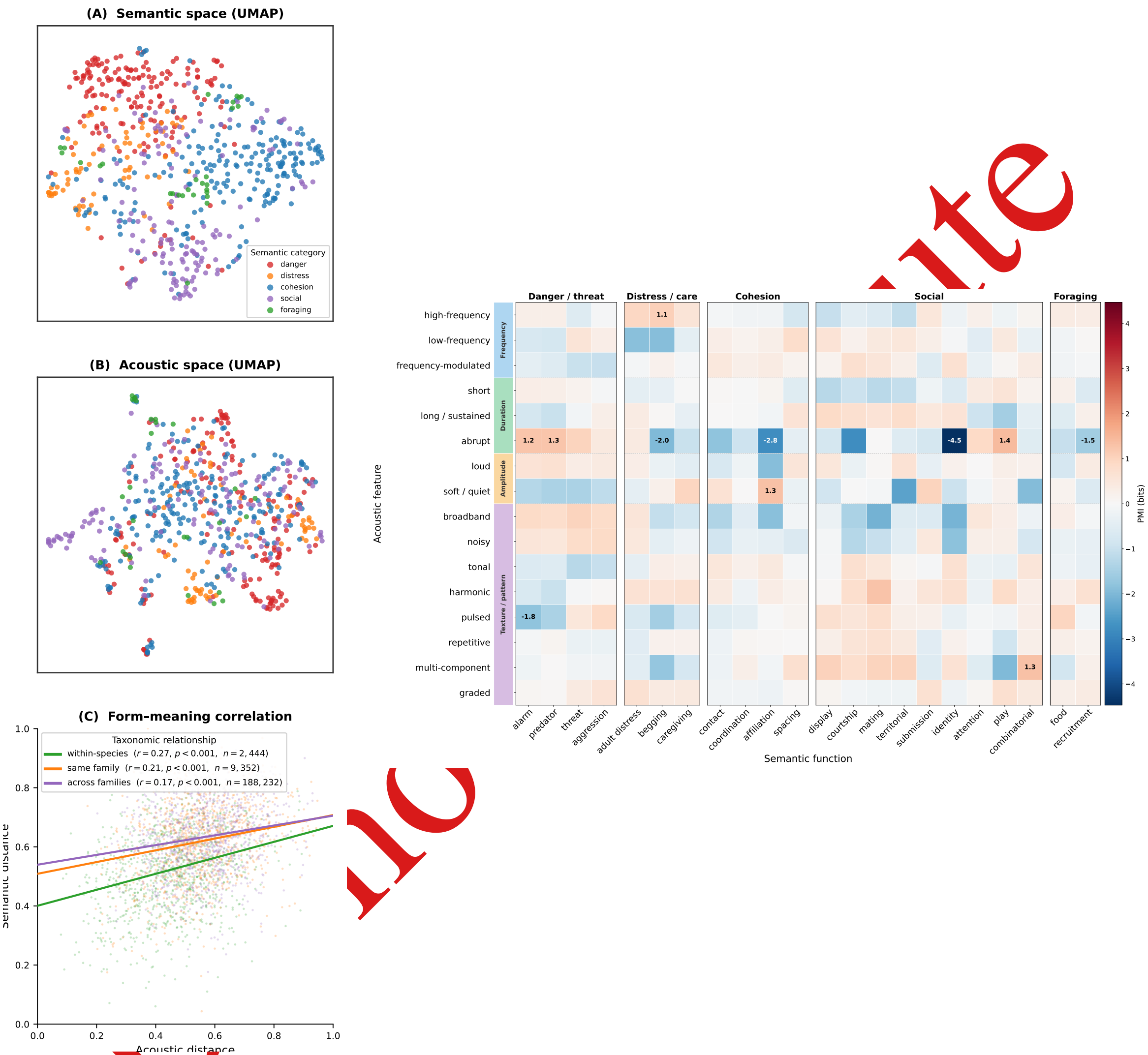

Figure 13. **[Draft] Merged Figs. 2+3 candidate.** Left column: **(A)** semantic UMAP, **(B)** acoustic UMAP, **(C)** Mantel scatter (same as Fig. 2). Right column: **(D)** PMI heatmap with semantic functions as rows and acoustic features as columns (square cells); rows and columns ordered by hierarchical clustering (same data as Fig. 4).

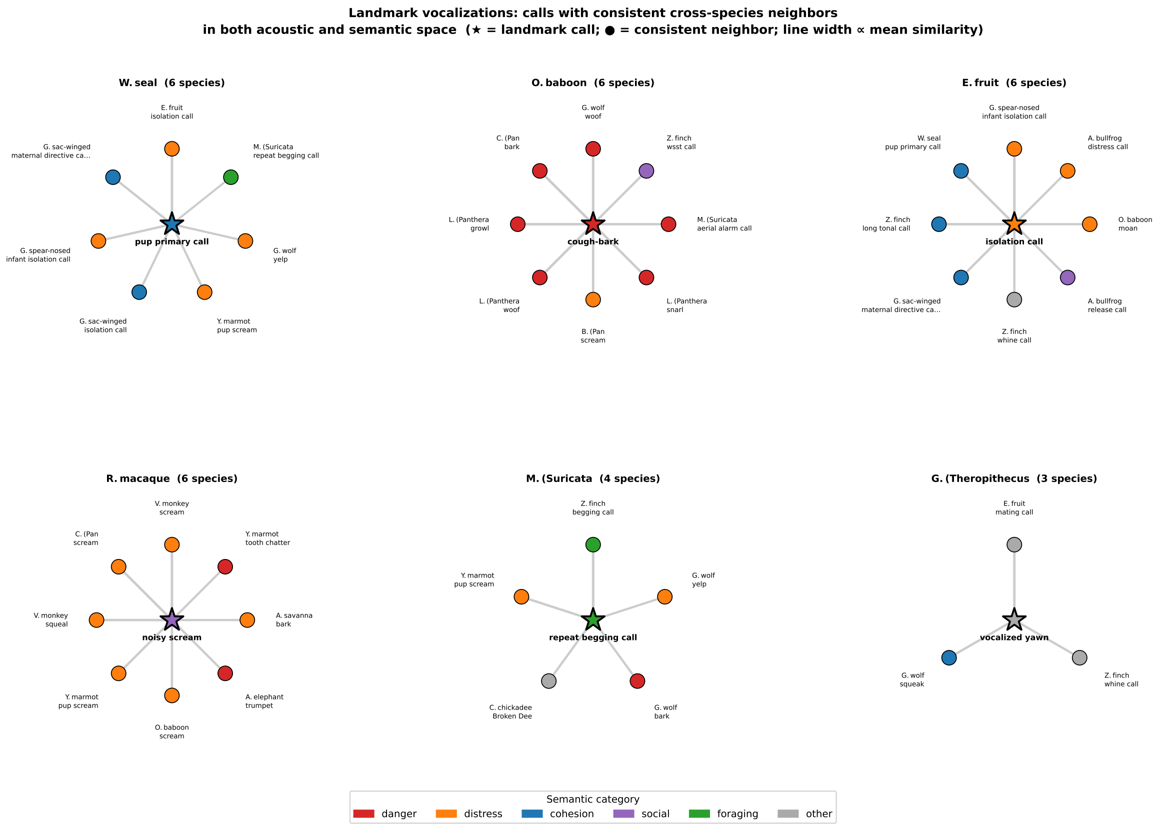

Figure 14. **[Alternative to Fig. 15] Cross-species lexical translation. Calls with consistent cross-species neighbors in both acoustic and semantic space.** Each rosette shows one landmark call (★, labeled with its call name at center) connected to up to eight calls from other species whose acoustic *and* semantic embeddings both rank among its top-20 nearest neighbors. Node color encodes semantic category (same scheme as Fig. 2); spoke width is proportional to mean acoustic–semantic similarity. Parenthetical counts give the number of distinct species with at least one consistent neighbor (range: 10–16). These landmark calls represent acoustic-semantic motifs independently co-opted across the tree of life—the concrete instances of the biological code quantified by the Mantel test (Fig. 2C).

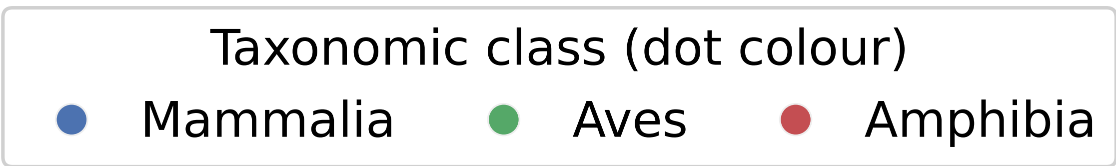

Figure 15. **Cross-species call groups sharing acoustic and semantic form.** Each panel shows one group of calls—one per species—belonging to a single semantic category (dark header). A pair of calls qualifies iff each is simultaneously the top-1 acoustic *and* top-1 semantic nearest neighbour of the other within its counterpart species; groups are cliques in this species-pairwise mutual-consistency graph, selected to maximise taxonomic diversity (class and family spread; see Supplementary Sections C and G). The coloured dot encodes taxonomic class (legend). *Top:* 2×2 layout; each row shows species and call name, followed by semantic (*Sem:*) and acoustic (*Acc:*) descriptions. *Bottom:* same four groups arranged horizontally; each row shows common name (line 1), call name (line 2), semantic keywords (line 3), and acoustic description in italics (line 4).